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Multi-scale modelling of flexible sandwich panels in large morphing structures

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Abstract

Morphing skin panels capable of undergoing large deformations are being investigated as a means of reducing fuel burn in next-generation aircraft through drag reduction. One approach to creating these panels is to 3D print zero Poisson's ratio cellular cores onto elastomeric facesheets, as proposed by the geometrically anisotropic thermoplastic rubber (GATOR) skins concept. Modelling these panels remains challenging due to multiple factors, including small feature sizes, hyperelastic material properties, geometric nonlinearities under large deformations, and complex interactions between the core and facesheets. While high-fidelity finite element (FE) analysis works well for unit cells and small components, it becomes prohibitively expensive for large-scale aircraft components. Hence, this work examines various approaches for mitigating the computational cost of analysing these sandwich panels, focusing on a morphing joint fairing for a folding wingtip. The analysis of these panels is simplified through scale separation, where the mechanical responses of the sandwich panel are homogenised into equivalent stiffness properties at the fine scale, allowing the structure to be modelled as a shell surface with equivalent stiffness properties at the coarse scale. This multiscale modelling approach can use homogenisation methods based on analytical formulations and FE analysis, as well as shell formulations such as Kirchhoff–Love and Reisner–Mindlin plate theory, each with simplifying assumptions that affect the accuracy of the results. This paper examines the effects of these approaches and their underlying assumptions on solution accuracy and computational cost, providing a comprehensive understanding of the modelling approaches available. The results indicate that the analytical homogenisation underpredicts the stiffness of the panel, the effects of transverse shear deformation become negligible for long panels, and the geometric nonlinearity in the panel deformation becomes significant for large rotations of the wingtip. The paper further presents an approach based on data-driven, second-order, multiscale modelling that captures the geometric nonlinearities in GATOR panels' responses accurately at a reduced computational cost.

1. Introduction

A sandwich panel consists of a core material stacked between two facesheets. Typically, facesheets are thin, stiff layers that provide rigidity to the sandwich panel through their in-plane stiffness. The core is usually relatively thick to increase the distance between the facesheets and the midplane, thereby increasing the panel's second moment of area and, therefore, bending rigidity. The core is stiff against thickness-wise deformation, maintaining the distance between the facesheets and the midplane. Sandwich cores also tend to be stiff in shear to effectively react to the shear loads generated between the upper and lower skins during bending. With this structure, sandwich panel architectures offer high out-of-plane stiffness at a low areal density, particularly when compared to isotropic plates. Additionally, the design partially decouples the in-plane and out-of-plane properties, which is desirable for various morphing structure applications, including in aircraft wings that adapt their shape to changing flight conditions [1, 2]. This partial decoupling of stiffness properties provides an approach to creating morphing skin panels that can achieve high in-plane flexibility while maintaining high out-of-plane stiffness [3, 4].

Flexible sandwich panels have been studied as adaptive skins for various wing morphing applications, including camber morphing [5], span extension [6, 7], wing twisting [8], and wingtip folding [9]. While the specific requirements of the skin panel may differ for each application, they all share the general morphing requirement of high out-of-plane stiffness and high in-plane flexibility in the morphing direction. High out-of-plane stiffness reduces the shape distortion of the wing due to aerodynamic loads and prevents the panel from buckling under compressive loads [5]. High flexibility in the morphing direction reduces the actuation energy required to achieve the shape change on the wing [5]. A zero Poisson's ratio panel also reduces the actuation energy by avoiding the additional energy required to restrain the panel deformation in the direction perpendicular to the morphing direction [6].

In morphing applications, flexible sandwich panels with cellular cores are preferred due to the extensive range of achievable equivalent elastic properties they offer through various core geometries [2]. These panels are made of repeating unit cells that tessellate to form the entire panel. Each unit cell typically consists of small features that require a fine mesh to accurately capture its deformation and reaction forces in a full-scale finite element (FE) analysis. Hence, full-scale FE analyses of flexible cellular core sandwich panels used in larger structures quickly become prohibitively expensive, especially from a design exploration and optimisation standpoint, where many different potential configurations would need to be simulated.

To reduce the computational cost, multiscale approaches can be used to analyse complex sandwich panels over large structures. In this approach, the mechanical response of the sandwich panel's unit cell is homogenised to equivalent shell stiffness properties. This equivalent shell stiffness is then used in a shell model that represents the full-scale three-dimensional geometry of the structure. This approach significantly reduces the computational cost of analysing the design space offered by complex sandwich panels in large morphing structures. Hence, this approach was employed by the authors in previous studies analysing flexible sandwich panels as a morphing fairing for folding wingtip joints [7, 8].

This paper presents a systematic study of the modelling assumptions that affect the accuracy and computational cost of the multiscale modelling approach in the context of a flexible sandwich panel fairing for folding wingtip joints. To this end, the following subsections first define this design problem, followed by a discussion of various modelling considerations for applying the multiscale modelling approach. This discussion is organised around three key considerations, namely:

(1) The type of equivalent stiffness properties required for the shell model of the fairing,

(2) The level of fidelity for the homogenisation method required to capture the sandwich panel's deformation mechanics effectively,

(3) The combined cost of the homogenisation method with the shell fairing model compared to a full-scale, full-fidelity analysis of the sandwich panel on the wing section.

1.1. Problem definition

One application that stands to benefit from the use of these morphing structures is the semi-aeroelastic hinge (SAH) concept [9, 10], shown in figure 1(a). This concept introduces a hinge joint between the wingtip and the inboard wing, with the hinge axis oriented at an angle to the aircraft's longitudinal axis. The hinge joint enables wingtip folding to shorten the wingspan, allowing the aircraft to use a smaller airport gate than would be possible without a hinge. The extended wingtip in flight provides a high aspect ratio wing, which reduces induced drag and, consequently, fuel consumption. The outward orientation of the hinge axis, combined with the release of the wingtip to respond freely to the airflow, alleviates the peak bending moment at the wing root during manoeuvres and gust encounters [11, 12]. Additionally, the free response of the wingtip alleviates the increased roll damping that would otherwise be caused by the longer wingspan, thereby aiding in achieving the required roll performance for certification [13]. Hence, SAH wingtips are expected to spend a non-negligible portion of the flight time in free response. Therefore, to realise this technology in commercial airliners, a fairing is desirable to cover the joint gaps and internal components to provide a smooth and continuous outer surface across the full range of wingtip folding angles.

The proposed fairing is a flexible sandwich panel with a carefully selected core geometry that enables high in-plane flexibility in the morphing direction while maintaining relatively high out-of-plane stiffness. The core features an accordion geometry [6, 14], also known as the MorphCore [15]. It has parallel ribs that provide high out-of-plane stiffness and a near-zero Poisson's ratio, and chevrons between the

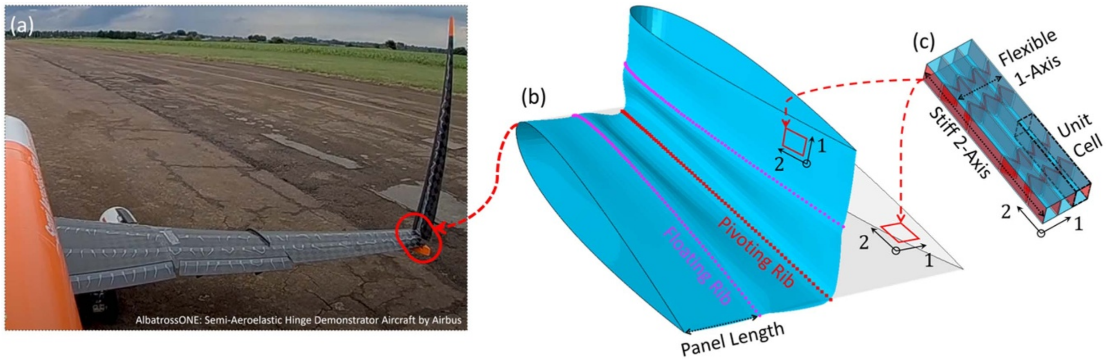

Figure 1. GATOR panel fairing on a folding wingtip joint [7]. (a) Shows the AlbatrossONE, a small-scale flying demonstrator for the semi-aeroelastic hinge (SAH) concept, flown by Airbus. Reproduced with permission from [10], (b) shows the proposed fairing section for the joint, along with the supporting ribs, and (c) shows the GATOR panel that makes up the fairing.

ribs that offer high in-plane flexibility in the morphing direction through their bending deformation [14, 16]. The MorphCore is sandwiched between two elastomeric facesheets, ensuring the edges of both the ribs and the chevrons are bonded to the facesheets, as shown in figure 1(c). This sandwich panel is fabricated through the fused fluid fabrication 3D printing technique with thermoplastic polyurethane (TPU), and is referred to as the geometrically anisotropic thermoplastic rubber (GATOR) sandwich panel [3, 4].

The GATOR panels are designed to cover the wingtip joint, as shown in figure 1(b), creating a smooth and continuous fairing that allows for deformation in the direction across the hinge (i.e. 1-axis) to fold the wingtip while transferring aerodynamic pressure loads to the supporting ribs [7]. This fairing provides a compliant surface that improves airflow around the joint and ensures an air- and water-tight seal to protect the joint components. The fairing is supported by a pivoting rib that is co-located with the wingtip joint but rotates independently of the wingtip folding angle [17]. The pivoting rib rotates to balance the reaction forces from the fairing, thereby redistributing the fairing deformation between the rib bays to reduce strain energy and, therefore, the torsional stiffness of the joint. Additional floating ribs have also been proposed to reduce the cross-section distortion of the fairing as the wingtip folds, but at the cost of increased torsional stiffness, which can be offset by increasing the span of the morphing fairing [7]. Pre-tension can also be applied to the fairing in the morphing direction to delay the compressive strain on the top surface and subsequent buckling of the panel.

As the fairing for the joint is a large structure, and the GATOR panel that constitutes the fairing has small features, high geometric nonlinearity and complex interaction between the core and facesheets [18], a careful consideration of the modelling techniques is required to achieve an accurate analysis with an affordable computational cost. The multiscale modelling approach is common in the study of repeating/architected structures, and various homogenisation approaches have been pursued in the literature, including analytical methods [6] and FE methods of different fidelities [19, 20]. The following sections examine the assumptions underlying various homogenisation schemes, their impact on the equivalent properties of the GATOR panel, and their implications for the multiscale modelling approach of the fairing.

1.2. Type of shell model for the fairing

The equivalent stiffness properties required to define the constitutive properties of shell elements depend on the shell formulation used in the fairing model. A laminated plate modelled using Kirchhoff–Love plate theory only requires the in-plane stiffness matrix (A), out-of-plane stiffness matrix (D) and the coupling matrix between the in-plane and the out-of-plane deformation (B) to define its constitutive properties. The formulation assumes that the plate's thickness remains unchanged and the cross-section remains straight and perpendicular to the mid-plane after deformation. These assumptions are valid only for thin plates with high transverse shear stiffness, where thickness-wise deformation and transverse shear deformations can be ignored.

The constitutive properties required for the Kirchhoff–Love shell formulation can be determined for the GATOR panel using either analytical homogenisation or FE-based second-order homogenisation techniques. The analytical approaches follow a two-step process, where the equivalent properties of the core are evaluated first, followed by the calculation of the equivalent shell stiffness matrix using the classical laminated plate theory (CLPT). In the FE-based approach, a unit cell of the GATOR panel is modelled using solid elements with appropriate periodic boundary conditions (PBC), and the unit cell is linearly perturbed in each plate deformation mode to evaluate the equivalent shell stiffness matrix. As the plate deformation modes include curvatures (i.e. bending modes), which are expressed in the solid elements as the second-order gradients of the displacement field, a second-order homogenisation technique is required to evaluate the equivalent shell stiffness [21]. Hence, the FE-based homogenisation technique discussed in this paper refers to the second-order, solid-to-shell homogenisation.

For plates with a high thickness-to-length ratio and low transverse shear stiffness, the transverse shear deformation of the plate becomes significant [22]. In this case, the Reissner–Mindlin plate theory must be employed, which extends the Kirchhoff–Love plate theory to include transverse shear deformation as an additional rotation of the straight cross-section. This approach requires an additional transverse shear stiffness matrix (**K**) along with the constitutive properties required for the Kirchhoff–Love plate. The true transverse shear strains of the plate in static bending are nearly parabolic across its thickness [19, 23] due to the traction-free boundary at the top and bottom surfaces of the plate. Hence, modelling transverse shear deformation simply as an additional rotation of the straight cross-section often overestimates the transverse shear stiffness in plates. Therefore, shear correction factors are required with the equivalent shear moduli in the transverse shear stiffness matrix to compensate for the additional artificial stiffness introduced to the shell model by the above-mentioned modelling assumption.

Analytical expressions for the shear correction factors are available for isotropic plates [24], orthotropic laminates [22] and solid-core sandwich panels [25]. However, to the authors' knowledge, general expressions applicable to sandwich panels with cellular cores are not available in the literature. Alternatively, an equivalent transverse shear stiffness matrix can be determined using an FE-based second-order homogenisation process. However, additional constraints are required on the unit cell to apply kinematically consistent transverse shear strains while keeping the top and bottom faces traction-free. An approach to applying these kinematically consistent transverse shear strains is to constrain the moment of fluctuation field stresses in the unit cell [19]. This approach eliminates the spurious stresses that would be generated in the unit cell when transverse shear deformation is applied using only the vertical faces. However, these constraints link all the nodes of the unit cell, making the homogenisation process computationally expensive for unit cells with fine meshes [19, 26]. Hence, this study does not explicitly evaluate the transverse shear stiffness matrix. Instead, the effects of ignoring the transverse shear stiffness are examined by comparing a model that ignores it with a model that accounts for it, as described in the following sections.

1.3. Deformation mechanics of the GATOR sandwich panel

A broad review of methods available in the literature for the parametrisation and homogenisation of cellular structures is presented by Thill *et al* [2], and Gibson and Ashby [27]. For brevity, this discussion will focus on methods directly relevant to the GATOR panel. The accuracy of the homogenised properties calculated by each homogenisation method depends on the fidelity with which the GATOR panel's deformation is modelled. In the analytical homogenisation, the equivalent in-plane properties of the core are evaluated by modelling the walls of the core as beams that deform in axial, bending, and transverse shear deformations [6, 27, 28]. The equivalent flexural moduli of the core significantly differ from the equivalent in-plane moduli due to the different loading conditions on the cell walls. The expressions evaluating the equivalent flexural moduli of the core require additional terms accounting for the torsional deformation of the walls [29]. A torsional coefficient is used in these expressions to account for thick cores, as the torsional deformation significantly differs between a beam-based model and a plate-based model of the core walls. This torsional coefficient is based on the height-to-length ratio of the wall and Poisson's ratio of the core material, and it approaches one as the height-to-length ratio of the wall approaches zero [30–32].

The closed-form expressions available for the torsional coefficient [32] significantly deviate from the FE-derived values as the height-to-length ratio of the wall increases; hence, tabulated values from FE analysis are typically used for the torsional coefficient [30]. Other approaches model the core walls as thin plates and solve them using the generalised variational principle, providing better agreement with the FE-based bending deformation of the cellular core [29]. However, these analytical techniques do not provide closed-form expressions that can readily evaluate the flexural moduli of a cellular core [31]. Hence, this study does not evaluate the equivalent flexural moduli of the core using the analytical homogenisation method, as an FE-based torsional coefficient is required to evaluate them accurately. Instead, the equivalent in-plane moduli of the core are used with the CLPT to evaluate the out-of-plane stiffness matrix of the GATOR panel. Considering that in a sandwich panel, the predominant contribution to the out-of-plane stiffness comes from the in-plane stiffness of the facesheets, this approach is not likely to introduce a significant error. However, the error introduced by this approach is identified by comparing its equivalent shell stiffness to the equivalent shell stiffness evaluated using FE-derived in-plane and out-of-plane moduli of the core with the CLPT.

The CLPT models a plate with multiple layers, assuming homogeneous elastic properties for each layer and perfect bonding between the layers. This approach implies that in a plate undergoing uniform deformation, the in-plane strains can be treated as uniform at any location at the same height (through the thickness) and continuous across the thickness of the plate. However, experimental and FE studies on the GATOR panel reveal a non-uniform strain distribution on the facesheets, where local strains can exceed the overall strain of the panel [18]. This non-uniform strain is attributed to the interaction between the chevrons in the core and the facesheet, resulting in the facesheet being pulled in the direction of the chevron displacement. The non-uniform strain on the facesheet increases the stiffness of the panel, thereby making the CLPTs estimate of the panel stiffness an underprediction. Hence, the effect of chevron-facesheet interaction is further studied in the following sections.

1.4. Solution accuracy and computational cost

The potential accuracy and computational cost of simulating a multiscale model depend on whether the constitutive properties of the shell elements are updated as the fairing deforms in order to capture geometric nonlinearities in the sandwich panel's response. As the GATOR panel deforms, the equivalent shell stiffness of the panel will change. An FE-based homogenisation can determine new equivalent shell properties for the GATOR panel unit cell. However, updating the new equivalent shell properties for each shell element requires a unit cell to be repetitively homogenised at the strain state of that element. In cases where the shell element has more than one integration point, the unit cell homogenisation is repeated for the strain state from each integration point. This repetitive homogenisation process for each integration point of the shell element, also known as a 'concurrent' approach, is computationally expensive (e.g. FE^2 method [19, 33]) and is not scalable for multi-scale modelling of large components.

Alternatively, a cheaper approach is to precompute the equivalent shell stiffness for various strain states of the unit cell and use interpolated values from the dataset to update the constitutive properties of the shell element as the analysis progresses. A precomputed dataset reduces computational cost compared to concurrent multiscale modelling, as it avoids the repetition of the homogenisation process for similar strain states. To identify the modelling fidelity required to accurately represent the GATOR panel fairing response using a shell-based simulation, this paper studies two multiscale modelling approaches. In the first approach, the GATOR panel's equivalent stiffness properties attained from the undeformed state of the unit cell are used as linear elastic stiffness properties for the shell-based fairing. In the second approach, the nonlinear elastic properties are modelled in the shell-based fairing by updating the constitutive properties for each element using the precomputed dataset of the GATOR panel's equivalent stiffness properties at various strain states. The errors introduced by each of these approaches are studied by comparing their results to the full-scale, full-fidelity model of the GATOR panel fairing.

1.5. Paper outline

This paper presents the effects of various homogenisation methods on the evaluated equivalent shell stiffness of the GATOR panel. These methods include an analytical method based on Timoshenko beam theory and CLPT, as well as a FE method that utilises second-order solid-to-shell homogenisation. The paper examines the significance of transverse shear deformation in the design problem by comparing the Kirchhoff–Love plate model with the Reissner–Mindlin plate model for a cantilevered GATOR panel subjected to tip displacement loading. It further compares the fairing responses evaluated using two different multiscale modelling approaches to those of a full-scale model, for a simplified twin-panel geometry representing a fairing slice along the span at the thickest chordwise location. One multiscale modelling approach uses equivalent shell stiffness properties of the GATOR panel from the undeformed state as linear elastic constitutive properties in the shell-based fairing model. The second multiscale modelling approach creates a data-driven surrogate model representing the equivalent shell stiffness for various strain states of the GATOR panel, and uses this surrogate model to update the constitutive properties of the shell elements in the fairing model.

The following sections are organised to present the effects of each of the following modelling approaches typically used in modelling cellular core-based sandwich panels on large structures.

(1) Timoshenko beam theory to model the core deformation,

(2) CLPT to model the GATOR panel stiffness,

(3) Kirchhoff–Love plate theory to model the GATOR panel deformation,

(4) Linear shell stiffness properties to model the GATOR panel fairing deformation,

(5) Nonlinear shell stiffness properties to model GATOR panel fairing deformation.

The methods used for analysis are presented in section 2. The first subsection describes the analytical homogenisation, and the second subsection describes the full-scale FE homogenisation of the sandwich panel unit cell. The third subsection describes the full-scale FE model used to study the effects of transverse shear deformation of the panel. The fourth subsection describes a full-scale model and a reduced-order shell model of a fairing slice aligned along the span at the thickest chordwise location. These models are used to study the assumptions that affect the accuracy of the analytical and reduced-order FE model by comparing it with higher-order models.

The results of the studies are presented in section 3. The first subsection assesses the error in the analytical models caused by ignoring localised stiffness at wall intersections. This paper proposes a revised formulation that reduces the effective length of the core walls. The second subsection assesses the error in GATOR panel models from ignoring the local nonuniform deformation of the facesheet. This paper indicates that FE-based homogenisation is required to capture the interaction between the facesheet and the core. The third subsection assesses the error from ignoring transverse shear deformation in sandwich panels for various panel lengths. This work quantifies the error for the GATOR panel in the target wing fairing problem. The fourth subsection assesses the error resulting from ignoring geometric nonlinearity in the modelling of large deformation. Hence, this paper highlights the need for alternative methods that achieve comparable accuracy to full-scale FE while scaling to large morphing structures. The final subsection presents a scalable, data-driven, multi-scale modelling approach that closely matches the full-scale FE model.

2. Methods

The multiscale modelling approach presented here focuses on the geometry of the fairing and the GATOR panel, shown in figure 1. The pivoting rib, whose rotation axis is co-located with the wingtip joint, maintains the distance between the joint and the fairing [17]. Additional floating ribs, attached only to the fairing, enforce the cross-section shape at their locations. These rib locations provide the reference unsupported length of the panel used in this study for validating the modelling approach. In the default configuration, the wing section has a root chord of 1.6 metres, representing the tip chord of the typical single-aisle jet airliner (e.g. A320 aircraft [34]). For simplicity, the aerofoil is assumed to be NACA 0015, giving a maximum cross-section thickness of 240 mm. The span of the morphing section is 0.8 m, giving an unsupported panel length of 200 mm between the inboard rib and the first floating rib. Note that in cases without floating ribs, the unsupported panel length is from the inboard rib to the pivoting rib, equalling 400 mm.

The geometry of the GATOR panel unit cell, along with the coordinate systems used, is shown in figure 2. Note that throughout this paper, two coordinate systems are used—the 123 axes to refer to the micro-scale stresses and strains, and the xyz axes to refer to the macro-scale stresses and strains. Additionally, the equivalent stiffness properties of the GATOR panel are also stated in terms of the 123 coordinate system. The figure shows the separation between the chevrons (d) defined at the tips of the chevrons. The distance between the base (i.e. chevron-rib intersection) of the adjacent chevrons is shown as cell rib length (h) and is defined as a function of other variables by equation (1).

$$h = 2l_c \sin\theta_c + t_c \sec\theta_c + d \tag{1}$$

This particular way of defining the core geometry ensures equal distance between the bases of the chevrons. Olympio and Gandhi [6] used this geometry to derive the analytical expressions for the equivalent properties of the core used in this study. However, this geometry results in the minimum distance between the chevrons pointing at each other (i.e. negative Poisson's ratio hexagonal cell) being smaller than the minimum distance between the chevrons pointing away from each other (i.e. conventional

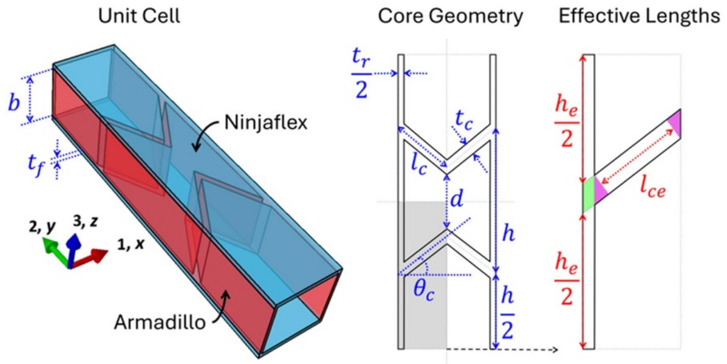

Figure 2. The geometry of the GATOR panel unit cell. Blue annotations indicate the variables that define the geometry. Red annotations indicate the effective lengths used in the modified analytical homogenisation of the core. Note that coordinate axes 123 refer to the micro-scale and axes *xyz* refer to the macro-scale.

Table 1. Dimensions of the unit cell geometry.

Variable		Default	Minimum	Maximum	Unit
Chevron angle	θ_c	60.0	40.0	70.0	deg
Chevron wall length	l_c	12.5	5.0	20.0	mm
Chevron wall thickness	t_c	1.0	0.5	1.5	mm
Chevron separation	d	6.0	2.0	10.0	mm
Rib thickness	t_r	1.0	0.5	1.5	mm
Core thickness	b	11.0	2.0	20.0	mm
Facesheet thickness	t_f	0.8	0.1	1.5	mm

hexagonal shape). The shorter distance between the chevrons pointing at each other results in a larger strain on the facesheet of the GATOR panel in this region [18]. Hence, this geometry definition is used in this paper solely for consistency with the analytical formulation—other studies by the authors on this concept employed different geometry definitions with equal spacing between the chevrons [7]. The default values for GATOR panel variables and their corresponding ranges, as shown in table 1, are based on a previous study by these authors [7], which identified suitable ranges of the design variables for this specific application.

In the default geometry configuration, the unit cell measures 12.5 × 59.3 × 12.6 mm in 1, 2 and 3 directions, respectively. These dimensions are referred to as length (L_1), width (L_2) and thickness (L_3), respectively. The multiscale modelling approach with first-order homogenisation typically requires the unit cell length scale to be at least an order of magnitude smaller than the panel length scale to keep the errors below 1% [35]. In contrast, these scale separation requirements are less stringent for second-order homogenisation [36]. With the default unit cell dimensions, 16 unit cells are required in the x direction to span the 200 mm length between the ribs for the case with one floating rib. Hence, this model has a characteristic length ratio of 16 between the fairing geometry and panel unit cell length scales, providing adequate scale separation for multiscale modelling of the GATOR panel as a shell-based fairing. Similarly, a length-to-thickness ratio above 15 is typically required for an isotropic plate to keep the transverse shear deformation negligibly small, enabling the use of Kirchhoff–Love plate formulation [37]. Although the GATOR panel has a length-to-thickness ratio of 15.9 for the unsupported length between the ribs, its behaviour will be significantly different from that of an isotropic panel. Hence, the length of the panel at which the transverse shear deformation can be ignored for the GATOR panel is studied further in the following sections.

GATOR panels are manufactured from elastomeric TPU materials with a stiffer formulation, Armadillo [38], used for the core, and a softer formulation, Ninjaflex [39], used for the facesheets. Both materials share similar chemical formulations, providing strong bonding between the core and the facesheet during 3D printing. The manufacturing process of GATOR panels and their mechanical characterisation through experimental testing has been presented by Heeb *et al* [3]. The panels were tested under axial deformation of up to 60% strain under cyclic loading, as well as under three-point bending deformation. GATOR panels show hysteresis behaviour under cyclic loading due to the hyperelastic nature of both materials. Hence, a hyperelastic material definition should ideally be used to model the GATOR panel. Another experimental study presented by Heeb and Woods analysed the deformation of

Table 2. Elastic properties of the materials used in the panel.

Property		Armadillo	Ninjaflex	Unit
Young's Modulus	E	396 [38]	22.9 [40]	MPa
Poisson's ratio	ν	0.48 [41]	0.48 [41]	

the panel under pressure loading using digital image correlation [18]. This study demonstrated the non-uniform strain distribution on the facesheet. Moreover, Heeb *et al* compared FE analysis of the GATOR panel using linear-elastic material properties with results from previous experimental studies. These comparisons showed good agreement between the FE model and the experimental results for the panel being stretched up to 10% in the morphing direction [40]. Hence, for simplicity, this study uses the linear elastic material properties for the materials, as shown in table 2.

The Young's modulus of Armadillo is from the datasheet provided by the manufacturer, NinjaTek [38], and the Young's modulus of Ninjaflex is from experimental data from the literature [40]. Poisson's ratio is not provided in the manufacturer's datasheet; hence, a typical value for TPU from the literature is used [41].

2.1. Analytical homogenisation

The analytical homogenisation of the GATOR panel's stiffness properties is carried out in two steps. First, the equivalent stiffness properties of the core are evaluated by considering the direct deformation of the unit cell walls. Second, the equivalent shell stiffness matrix for the sandwich panel is evaluated using the CLPT.

The analytical expressions for the equivalent elastic moduli of MorphCore [14], also known as the accordion geometry [6, 16], were derived by Olympio and Gandhi [6] for the geometry defined with blue annotations in figure 2. These expressions are based on the direct deformation of the unit cell walls in axial, bending, and transverse shear deformation. However, they tend to underpredict the core's equivalent moduli as the bending members' lengths (e.g. chevron and rib walls) that actively participate in deformation are shorter than their full lengths—due to the stiffening effect of the interaction area between the chevrons and the rib walls, and between the two halves of the chevron at their intersection. Hence, the expressions from Olympio and Gandhi [6] are modified here by assuming the interaction areas are effectively rigid, leading to a reduction in the length of the wall undergoing deformation, as shown with red annotations in figure 2. These effective lengths of the chevron wall and cell rib wall are given by equations (2) and (3), respectively.

$$l_{ce} = l_c - \frac{t_r}{2\cos\theta_c} - t_c \tan\theta_c \tag{2}$$

$$h_e = h - \frac{t_c}{\cos\theta_c}. \tag{3}$$

The modified expressions for the equivalent moduli of the core using the effective lengths are shown in equations (4)–(6).

$$\frac{E_1}{E} = \frac{{\beta_e}^3}{\alpha\cos\theta_c\left({\beta_e}^2 + \tan^2\theta\ \left(1+\kappa{\beta_e}^2\right)\right)} \tag{4}$$

$$\frac{E_2}{E} = \frac{\alpha\,\eta\,\beta_e}{2\alpha_e\ \cos\theta_c} \tag{5}$$

$$\frac{G_{12}}{E} = \frac{{\beta_e}^3\cos\theta_c}{\frac{2}{\alpha}\left(\frac{\alpha_e}{\eta}\right)^3\ \cos^2\theta_c\ \left(1+\kappa\left(\frac{\beta_e\,\eta}{\alpha_e}\right)^2\right) + \alpha\ \left(4\ \cos^2\theta_c\ \left(1+\kappa\ {\beta_e}^2\right) + {\beta_e}^2\sin^2\theta_c\right)} \tag{6}$$

where:

$$\eta = \frac{t_r}{t_c},\quad \beta_e = \frac{t_c}{l_{ce}},\quad \alpha_e = \frac{h_e}{l_{ce}},\quad \alpha = \frac{h}{l_c}.$$

The parameters E and ν are Young's modulus and Poisson's ratio of the core material, respectively. The parameter κ is a correction factor used with the transverse shear deformation [27], and it is evaluated as $\kappa = 2.4 + 1.5\,\nu$. Note that these expressions are identical to that of Olympio and Gandhi [11] for the case where the effective lengths of the chevron and the cell rib equal their full lengths (i.e. $h_e = h$ and $l_{ce} = l_c$).

The Poisson's ratio (υ_{xy}) of the core is assumed to be zero due to the ribs restricting deformation in the 2-axis. The modified expressions in equations (4)–(6) provide the in-plane properties of the core required for evaluating the shell stiffness matrix using the CLPT. The shell stiffness matrix relates the strains and curvatures of the panel to the distributed forces and moments along the edges of the plate according to equation (7).

$$\begin{bmatrix} \boldsymbol{N} \\ \boldsymbol{M} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{D} \end{bmatrix} \begin{bmatrix} \boldsymbol{\varepsilon} \\ \boldsymbol{\kappa} \end{bmatrix} \tag{7}$$

where:

$$\boldsymbol{N} = \begin{bmatrix} N_{xx} & N_{yy} & N_{xy} \end{bmatrix}^T, \quad \boldsymbol{M} = \begin{bmatrix} M_{xx} & M_{yy} & M_{xy} \end{bmatrix}^T,$$

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx}\varepsilon_{yy}\gamma_{xy} \end{bmatrix}^T, \quad \boldsymbol{\kappa} = \begin{bmatrix} \kappa_{xx} & \kappa_{yy} & \kappa_{xy} \end{bmatrix}^T.$$

The vector $\boldsymbol{N}$ is the forces, $\boldsymbol{M}$ the moments, $\boldsymbol{\varepsilon}$ the strains, and $\boldsymbol{\kappa}$ the curvatures. For clarity, the combined vector of distributed forces and moments is hereafter referred to as distributed loads, and the combined vector of strains and curvatures is referred to as the strain state. The shear strain and torsional curvature are defined in the engineering convention, where $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$ and $\kappa_{xy} = 2\,\partial^2 w/\partial y \partial x$, respectively. The sub-matrices of the shell stiffness matrix are evaluated as shown in equations (8)–(10).

$$A_{ij} = \int_{-t_p/2}^{t_p/2} Q_{ij}\,\mathrm{d}z \tag{8}$$

$$B_{ij} = \int_{-t_p/2}^{t_p/2} Q_{ij}\,z\,\mathrm{d}z \tag{9}$$

$$D_{ij} = \int_{-t_p/2}^{t_p/2} Q_{ij}\,z^2\,\mathrm{d}z \tag{10}$$

where:

$$\boldsymbol{Q} = \frac{1}{(1-\nu_{12}\nu_{21})} \begin{bmatrix} E_1 & \nu_{21}E_1 & 0 \\ \nu_{12}E_2 & E_2 & 0 \\ 0 & 0 & G_{12}\,(1-\nu_{12}\nu_{21}) \end{bmatrix}.$$

The subscripts i and j are the matrix indices, and the variable t_p is the thickness of the whole panel. The matrix $\boldsymbol{Q}$ represents the in-plane stiffness of each layer normalised by the thickness. The $\boldsymbol{Q}$ matrix for the core is evaluated using the equivalent properties from equation (4) to equation (6). The $\boldsymbol{Q}$ matrix for the facesheet is evaluated using the isotropic properties of Ninjaflex from table 2. The equivalent shell stiffness matrix of the GATOR panel, as determined by equations (8)–(10), can be used to define the constitutive properties of the shell elements for the multiscale analysis of the fairing.

The expressions used to evaluate the equivalent moduli of the core assume that the walls of the core can be modelled as Timoshenko beams. The CLPT used to evaluate the equivalent shell stiffness of the GATOR panel assumes uniform strains at any location at the same height (through the thickness) from the midplane. The errors introduced by these modelling assumptions are studied in sections 3.1 and 3.2 by comparing the results from analytical and FE-based homogenisation methods.

2.2. FE homogenisation

The unit cell of the GATOR panel, also referred to as the representative volume element (RVE), is modelled using solid elements, and its constitutive response is evaluated as a solution to a classical boundary value problem [21]. This approach captures the effects of material heterogeneity and non-uniform strains within the RVE. As the fairing geometry is modelled using shell elements, a solid-to-shell homogenisation is required on the RVE model to evaluate the equivalent shell stiffness for the fairing model. Note that the shell elements in the fairing model exhibit higher-order deformation modes, such as bending curvature, which are expressed in the solid elements of the RVE as second-order gradients of the displacement field [21]. Hence, a second-order homogenisation method is required to determine the equivalent shell stiffness of the RVE. This section describes the implementation of a second-order, solid-to-shell homogenisation approach for this study.

In the FE-based homogenisation, the RVE represents a repeating unit of the panel that tessellates to form an infinite panel. Hence, the deformation of each RVE face must be compatible with the deformation of the opposite face at the adjacent RVE. PBC are applied to the RVE face nodes to satisfy this deformation compatibility condition. The PBCs couple the identical nodes on opposite faces of the RVE through a set of constraint equations [20, 42, 43]. Readers are directed to the [20] for the derivation of these constraints for second-order solid-to-shell homogenisation. Here, these constraints are stated as nodal equations, along with additional terms corresponding to external nodes, which are used to apply the macro-scale strains to the RVE, as shown in equations (11)–(13).

$$-u_1^1 + u_1^2 + Au_1^{\varepsilon} + Bu_1^{\kappa} + \frac{1}{2}Cu_3^{\varepsilon} + \frac{1}{2}Du_3^{\kappa} = 0 \tag{11}$$

$$-u_2^1 + u_2^2 + \frac{1}{2}Au_3^{\varepsilon} + \frac{1}{2}Bu_3^{\kappa} + Cu_1^{\varepsilon} + Du_1^{\kappa} = 0 \tag{12}$$

$$-u_3^1 + u_3^2 + Eu_1^{\kappa} + Fu_2^{\kappa} + \frac{1}{2}Gu_3^{\kappa} = 0 \tag{13}$$

where:

$$\begin{array}{l} A = X_1^1 - X_1^2, \quad B = zA, \quad C = X_2^1 - X_2^2, \quad D = zC \\ E = -\frac{1}{2}\left(\left(X_1^1\right)^2 - \left(X_1^2\right)^2\right), \quad F = -\frac{1}{2}\left(\left(X_2^1\right)^2 - \left(X_2^2\right)^2\right) \\ G = -\left(X_1^1 X_2^1 - X_1^2 X_2^2\right) \end{array} .$$

The displacements and the coordinates of each node are annotated as u_D^f and X_D^f, respectively, where the subscript D refers to the degree of freedom, and the superscript f is the face identifier, with 1 and 2 referring to the positive and negative face of each axis. The constraint equations define the relative displacement of the node pair in each degree of freedom. The constraint equations couple the macro-scale strains ε_{xx}, ε_{yy} and γ_{xy} of the RVE to the 1, 2, and 3 direction displacements of the external node u_D^{ε}, respectively. Similarly, the constraint equations also couple the macro-scale curvatures κ_{xx}, κ_{yy} and κ_{xy} of the RVE to the 1, 2, and 3 direction displacements of the external node u_D^{κ}. The locations of the external nodes are arbitrary since only their applied displacements affect the RVE deformation. A factor $1/2$ is used with the coefficient of shear and torsion terms in the constraint equations, as the shear strain and torsional curvature are defined in the engineering convention.

The equations can be intuitively explained for an individual node pair. For instance, consider a node pair on the 1-axis, as shown in figure 3.

As the unit cell is centred about the origin, the 2-axis and 3-axis coordinates of the pair will be identical (i.e. $X_2^1 = X_2^2$ and $X_3^1 = X_3^2$ in the nodal coordinate notation). However, the 1-axis coordinates of the pair will have the same magnitude with opposite signs (i.e. $X_1^1 = L_1/2$ and $X_1^2 = -L_1/2$). For this case, the constraint equations can be simplified as equations (14)–(16).

$$-u_1^1 + u_1^2 + L_1 u_1^{\varepsilon} + zL_1 u_1^{\kappa} = 0 \tag{14}$$

$$-u_2^1 + u_2^2 + \frac{1}{2}L_1 u_3^{\varepsilon} + \frac{1}{2}zL_1 u_3^{\kappa} = 0 \tag{15}$$

$$-u_3^1 + u_3^2 - \frac{1}{2}yL_1 u_3^{\kappa} = 0 \tag{16}$$

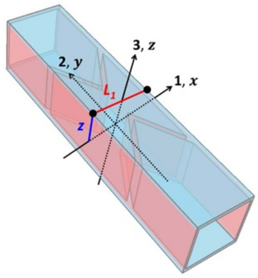

Figure 3. Example of a node pair for periodic boundary conditions. The node at the face on the positive side of x -axis has the coordinates $X_1^1 = L_1/2, X_2^1 = 0$ and $X_3^1 = 0$. The node at the face on the negative side of x -axis has the coordinates $X_1^2 = -L_1/2, X_2^2 = 0$ and $X_3^2 = 0$. Note that coordinate axes 123 refer to the micro-scale and axes xyz refer to the macro-scale.

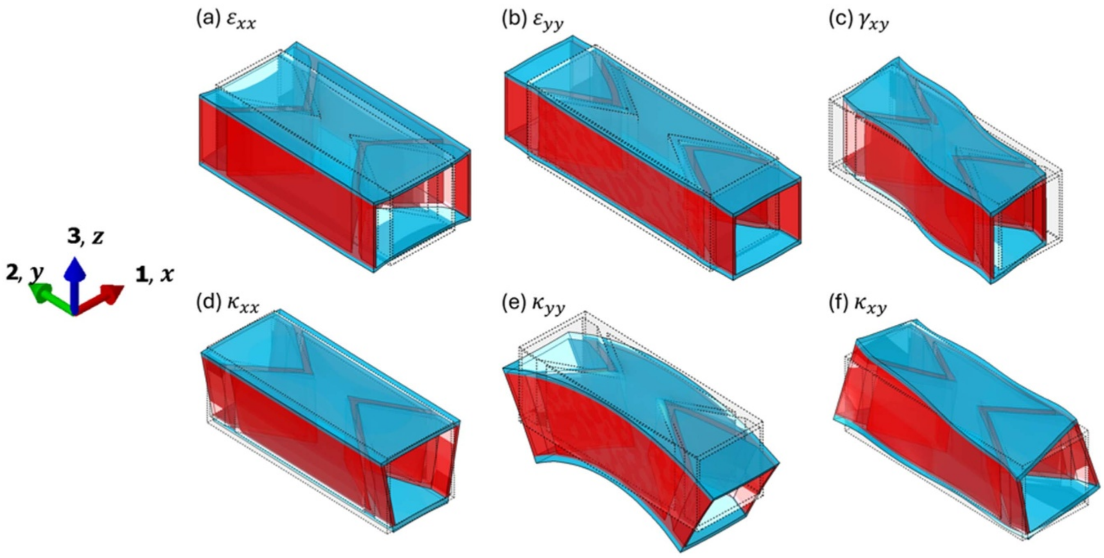

Figure 4. GATOR panel unit cell deformed to each shell deformation mode. Note that coordinate axes 123 refer to the micro-scale and axes xyz refer to the macro-scale.

where:

$$z = X_3^1 = X_3^2, \quad y = X_2^1 = X_2^2.$$

These expressions indicate that for a node pair on the 1-axis:

- The relative displacement in 1-direction is only due to macro-scale strain and curvature in the x-axis (i.e. in micro-scale 1-direction),
- The relative displacement in 2-direction is only due to macro-scale shear and torsion,
- The relative displacement in 3-direction is only due to macro-scale torsion.

As the PBCs define the difference in the node pair displacement in each degree of freedom, a case with zero relative displacements can still have non-zero individual displacements as long as they are identical on both nodes. For instance, in pure bending of the RVE along the 1-direction, both nodes are individually displaced in the 3-direction with identical displacement, so the difference between them is zero.

The homogenisation procedure is carried out by loading the RVE in each shell deformation mode and evaluating the reaction forces on the external nodes used to apply the deformations. The deformed shapes of the GATOR panel unit cell in the shell deformation modes are shown in figure 4.

For instance, consider applying a pure strain of $\varepsilon_{xx} = 1$, which required the applied displacement on the reference nodes to be $u_1^\varepsilon = 1$ and $u_2^\varepsilon = u_3^\varepsilon = u_1^\kappa = u_2^\kappa = u_3^\kappa = 0$. The column of the shell stiffness matrix corresponding to each applied deformation is given by equation (17), where A_{RVE} refers to the midplane area of the RVE, and the index i refers to the row of the applied shell deformation mode and the column of the stiffness matrix corresponding to that applied deformation.

$$\boldsymbol{K}_{plate}(:,i) = \frac{1}{u_i A_{RVE}} \boldsymbol{F} \tag{17}$$

where:

$$\boldsymbol{K}_{plate} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{D} \end{bmatrix},$$
$$\boldsymbol{F} = F_1^{\varepsilon}, F_2^{\varepsilon}, F_3^{\varepsilon}, F_1^{\kappa}, F_2^{\kappa}, F_3^{\kappa}]^T, \quad \boldsymbol{u} = [u_1^{\varepsilon}, u_2^{\varepsilon}, u_3^{\varepsilon}, u_1^{\kappa}, u_2^{\kappa}, u_3^{\kappa}]^T.$$

Hence, for the case of $\varepsilon_{xx} = 1$, the index $i = 1$, giving the first column of the stiffness matrix. This procedure requires six load cases, each corresponding to a shell deformation mode, to fully populate the shell stiffness matrix. Each load case is analysed using a linear perturbation step in Abaqus, evaluating the equivalent stiffness from the tangential stiffness of the RVE at its initial strain state. The RVE is homogenised at its undeformed state to define linear shell stiffness properties for the shell-based fairing model. For cases with nonlinear shell stiffness properties, the dataset of equivalent shell stiffness at various strain states is evaluated by first loading the RVE to a specific macro-scale strain state and then repeating the homogenisation procedure.

The FE models for homogenisation and full-scale analysis use first-order brick elements with reduced integration (C3D8R). The FE analysis is carried out using Abaqus [37], a commercial FEM solver. The meshes for the FE analysis are generated using GMSH [44], an open-source software for design and meshing. The process of model generation, meshing, analysis and results extraction is automated using bespoke Python scripts, as both GMSH and Abaqus allow access to them through an application programming interface (API).

The GATOR panel models are meshed with 1 mm elements to give equivalent moduli values converged to within ±1% error compared to models with half the element size. Similarly, models of the core are meshed with 0.25 mm elements, giving equivalent moduli values that are converged to within ±1% error compared to models with half the element size. The finer mesh required for the core models is due to the sensitivity of the equivalent E_x value to the mesh size. However, as the contribution of the facesheets to the equivalent moduli is more significant in the GATOR panels, a coarse mesh was adequate to get converged results for the GATOR panels.

In sections 3.1 and 3.2, the results from the FE-based homogenisation procedure are compared with those from the analytical approach to identify the effects of assuming the Timoshenko beam model for the deformation of the core walls and CLPT for evaluating the equivalent shell stiffness. The FE-based homogenisation procedure used does not evaluate the equivalent transverse shear stiffness of the RVE. Hence, the effect of transverse shear deformation on the flexural stiffness of the panel is studied next by comparing the equivalent flexural stiffness from FE-based homogenisation with the equivalent flexural stiffness evaluated from a full-scale FE model of a cantilevered GATOR panel deformed with a tip displacement.

2.3. Full-scale panel analysis

Evaluation of the equivalent transverse shear stiffness matrix requires a kinematically consistent second-order solid-to-shell homogenisation process, which can apply pure transverse shear strain to the RVE while keeping the top and bottom faces traction-free. As highlighted in section 1.2, this is achieved by additional constraints applied to eliminate the spurious stresses generated within the RVE, which typically involves constraining all the nodes of the RVE [19]. Hence, evaluating the transverse shear stiffness matrix makes the homogenisation process expensive and difficult to scale for RVEs with a large number of degrees of freedom (e.g. an RVE with a very fine mesh or multiple unit cells). Therefore, this study investigates the error introduced in panels of various lengths by ignoring transverse shear deformation to assess whether this simplification can be applied in this particular application.

In this study, the displacement and reaction load data from a full-scale FE simulation of the GATOR panel are used in two analytical expressions for evaluating an equivalent flexural modulus (E_1^D). These analytical expressions are derived from Kirchhoff–Love and Reissner–Mindlin plate formulations. The equivalent flexural modulus values evaluated from these expressions are compared to those from the homogenisation process (i.e. D-matrix) to identify the effects of transverse shear deformation.

For this study, GATOR panels of various lengths are modelled with a cantilever fixture on one end and a vertical displacement applied at the other, as shown in figure 5.

The dimensions of the panel are referred to as length (L_x), width (L_y) and thickness (L_z) in x, y and z directions, respectively. Various panel lengths along the x-axis, ranging from 1 cell to 20 cells, are used

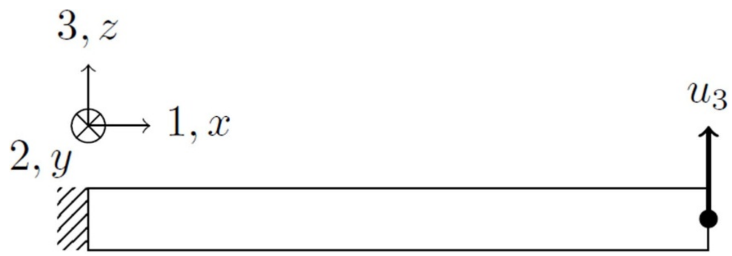

Figure 5. Tip displacement model of the GATOR panel used to evaluate equivalent flexural modulus and study the effects of transverse shear. Note that coordinate axes 123 is used to define equivalent stiffness properties (i.e. micro-scale) and axes *xyz* is used to define the mechanics of the structure (i.e. macro-scale).

to investigate the significance of transverse shear deformations in panels of various lengths. Only a single cell width is used along the y-axis, since symmetric boundary conditions are applied to the nodes on the front and back faces (i.e. faces normal to the y-axis) to eliminate free-edge effects as the panel deforms. The nodes at the left face are fixed, and the nodes on the right face are rigidly connected to the midpoint of the face. A vertical displacement (u_3) is applied to this midpoint at the tip, and its vertical reaction force (F_3) and the rotation angle (θ_2) about the y -axis are measured at the end of the simulation. The FE analysis is carried out using a linear step in the Abaqus FE solver [37].

The governing moment equations from the Kirchhoff–Love and Reissner–Mindlin plate models are shown in equations (18) and (19), respectively.

$$\text{Kirchhoff} - \text{Love Plate}\;:\; M_{xx} = \frac{E_1^D I}{1 - \upsilon_{xy}\upsilon_{yx}} \left(\frac{\mathrm{d}^2 w}{\mathrm{d}x^2} + \upsilon_{yx} \frac{\mathrm{d}^2 w}{\mathrm{d}y^2} \right) \tag{18}$$

$$\text{Reissner} - \text{Mindlin Plate}\;:\; M_{xx} = \frac{E_1^D I}{1 - \upsilon_{xy}\upsilon_{yx}} \left(\frac{\mathrm{d}\phi_x}{\mathrm{d}x} + \upsilon_{yx} \frac{\mathrm{d}\phi_y}{\mathrm{d}x} \right). \tag{19}$$

These equations are used to derive the expressions for the equivalent flexural modulus (E_1^D) of the panel. The vertical displacement (w) represents the z-axis displacement of the midplane. The distributed moment (M_{xx}) and rotation angle (ϕ_x) are defined about the y-axis (at the cross-section face normal to the x-axis). The rotation angle in the Reissner–Mindlin plate model includes cross-sectional rotation resulting from both bending and transverse shear deformation. In contrast, in the Kirchhoff–Love model, the rotation of the cross-section is only due to bending. As the distributed moment (M_{xx}) represents the moment normalised by its respective edge length, the second moment of area (I) is also normalised by its edge length, thereby making it a function of only the panel thickness (i.e. $I = L_z^3/12$).

The symmetric boundary conditions at the front and rear faces restrict the panel deformation to a monoclastic curvature along its length (x-axis), thereby simplifying the plate equations to a 1-dimentional problem (due to $d^2w/dy^2 = d\phi_y/dx = 0$). Hence, after integrating the plate equations along its length, the expressions derived for the equivalent flexural modulus from Kirchhoff–Love and Reissner–Mindlin plate models are shown in equations (20) and (21), respectively.

$$\text{Kirchhoff} - \text{Love Plate}\;:\; E_1^D = 4\frac{F_3}{u_3}\frac{1}{L_y}\left(\frac{L_x}{L_z}\right)^3 \times \left(1 - \upsilon_{xy}\upsilon_{yx}\right) \tag{20}$$

$$\text{Reissner} - \text{Mindlin Plate}\;:\; E_1^D = 6\frac{F_3}{\phi_2}\frac{1}{L_y}L_x^2\left(\frac{1}{L_z}\right)^3 \times \left(1 - \upsilon_{xy}\upsilon_{yx}\right). \tag{21}$$

The Poisson's ratios in these equations are evaluated from the out-of-plane stiffness matrix (i.e. D-matrix) obtained through the FE-based homogenisation.

As the Kirchhoff plate model does not account for the transverse shear deformation, the displacement due to transverse shear in the full-scale FE model is seen by the analytical expression as displacement due to bending, leading to an underprediction of the equivalent flexural modulus. Hence, the error in the equivalent flexural modulus from the Kirchhoff–Love plate is expected to be large for short panels where the transverse shear deformation is significant. This error is expected to decrease for longer panels as the transverse shear deformation becomes negligible for panels with a high length-to-thickness ratio (typically greater than 15 for isotropic panels [37]). In contrast, as the Reissner–Mindlin plate model

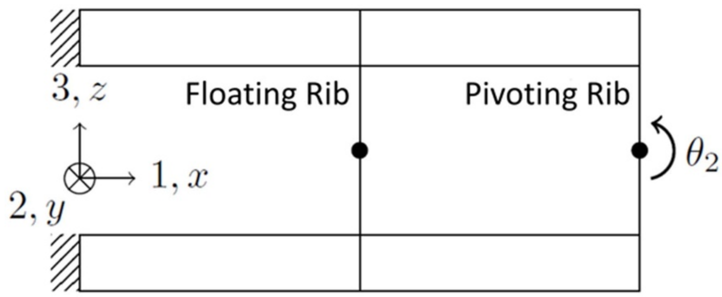

Figure 6. Representative fairing model for analysis of the effects of geometric nonlinearity in the GATOR panel deformation. Note that coordinate axes 123 is used to define equivalent stiffness properties (i.e. micro-scale) and axes *xyz* is used to define the mechanics of the structure (i.e. macro-scale).

accounts for transverse shear deformation, the error in equivalent flexural modulus is expected to be small for all panel lengths.

This study aims to determine whether the transverse shear effects become negligible for panels shorter than the unsupported length of the fairing between the ribs (i.e. length of 16 cells). If the error in flexural modulus from the Kirchhoff–Love plate model becomes negligible for panel lengths shorter than 16 cells, then a shell model based on the Kirchhoff–Love plate formulation is adequate to model the fairing in the multiscale analysis. In contrast, if large errors exist in the equivalent flexural modulus from the Kirchhoff–Love plate model for panel lengths beyond 16 cells, then the Reissner–Mindlin plate formulation must be used in the multiscale modelling approach, along with a second-order solid-to-shell homogenisation method that can evaluate the transverse shear stiffness matrix. The results of these analyses are presented in section 3.3.

2.4. Simplified fairing analysis

The shell-based modelling approach used to analyse the fairing in previous studies [7, 8] and in this section assumes linear elastic constitutive properties. These constitutive properties are defined using the homogenised shell stiffness of the GATOR panel from the undeformed state. Previous high-fidelity studies of the GATOR panel, from FE simulations and experimental testing, have demonstrated geometric nonlinearity in its deformation, resulting in nonlinear stiffness properties [40]. Hence, the effects of ignoring the nonlinearity in the equivalent stiffness are studied by comparing the response of this shell model with linear stiffness properties to that of the full-scale model, which captures the geometric nonlinearity in the GATOR panel deformation.

To make the full-scale analysis computationally affordable, a simplified fairing section is used in this study with two flat GATOR panels representing a fairing slice along the span from the inboard rib to the pivoting rib at the thickest chordwise location for a typical jet airliner wing section, as shown in figure 6.

Only the inboard half of the fairing is modelled (hence the presence of the wingtip hinge, pivoting rib, and applied rotation θ_2 at the rightmost end of the model) due to the symmetry of the fairing in this simplified model. The full-scale model employs first-order brick elements with reduced integration (C3D8R) and a mesh size of 1 mm, as detailed in section 2.2. The shell model employs first-order quadrilateral elements with reduced integration (S4R) and a mesh size of 8 mm, yielding converged results in torque values compared to models with half its element size.

The spacing between the ribs is the same as the original wing section, and the thickness (i.e. height in 3-axis) of this representative fairing section is the maximum thickness of the wing section. For NACA 0015 aerofoil section with a 1.6 m chord, the maximum thickness is 240 mm. Hence, given that the GATOR panel thickness is 12.6 mm, the multiscale model has two shell surfaces with 227.4 mm between them. For both the full-scale and the shell models, the left end is fixed, and the nodes of the ribs are rigidly connected to their midpoints (shown as a dot on each rib in figure 6). The midpoint of the pivoting rib is constrained in translation in 2 and 3 axes and in rotation in 1 and 3 axes. As the wing section fairing is envisioned to have span-wise pre-tension [7, 17], a pre-strain is applied to this representative model by displacing the pivoting rib's midpoint in the 1-axis direction. In the case without pre-strain, the midpoint of the pivoting rib is also constrained in translation in the 1-axis. A rotation is applied to the midpoint of the pivoting rib about the 2-axis, and the reaction torque on this node is measured to study the change in torsional stiffness due to deformation. The torsional stiffness in this

Table 3. Ranges of the macro-scale strain values applied to the unit cell.

Strain component	Minimum	Maximum
ϵ_{xx}	-20%	$+40\%$
ϵ_{yy}	$-\ 2\%$	$+2\%$
ϵ_{xy}	-15%	$+15\%$
κ_{xx}	-1.5%	$+1.5\%$
κ_{yy}	-1.5%	$+1.5\%$
κ_{xy}	-0.5%	$+0.5\%$

case is simply the ratio of the incremental change in the reaction moment to the incremental change in the applied rotation angle.

The torque-rotation responses of the linear elastic shell model and the full-scale model for the twin-panel are compared to evaluate the accuracy of the modelling approach. Moreover, variations in two design variables—namely, the number of floating ribs and the pre-strain on the fairing—are used to study the robustness of the agreement between the linear elastic shell model and full-scale models. It is expected that the full-scale model will exhibit a softening of torsional stiffness as the pivoting rib rotates due to the buckling of the facesheets of the top GATOR panel, which is being compressed. The local buckling of the facesheets is not captured in the shell model with linear stiffness; hence, its torque result is expected to deviate from the full-scale model's torque values following the onset of facesheet buckling. While pre-strain is expected to delay the onset of facesheet buckling, it is also expected to cause changes in the equivalent stiffness of the GATOR panel. The comparison of torque results from the linear elastic shell model and the full-scale model is used to identify the limits of the maximum folding angle and the ranges of pre-strain where the linear elastic shell model is adequate to study the fairing response. The results of these analyses are presented in section 3.4.

2.5. Data-driven multiscale modelling method

The classical concurrent multiscale modelling approach computes and updates the constitutive properties at each material point through a nested analysis and homogenisation of the underlying unit cell. This approach accurately bridges the nonlinear behaviour at the material length scale to that at a coarser length scale [36]. However, as previously highlighted, it is prohibitively expensive and does not scale for large components. This bottleneck can be addressed through a data-driven approach [45–47] by pre-computing a material response database and then fitting it with a surrogate model that is emulated at runtime, effectively replacing the 'online' analysis of the unit cells. This section presents a multiscale modelling approach that captures the nonlinearity in the GATOR panel stiffness as it deforms using a data-driven surrogate model to update the constitutive properties of the shell-based fairing.

For a Kirchhoff–Love shell problem, owing to symmetry, the database will contain 18 unique stiffness components of the ***ABD*** matrices mapped from 6 unique strain components (i.e. 3 membrane and 3 curvature components). The sampling strategy needs to strike a balance between available computer resources and sufficient coverage of the strain space to ensure the material model is suitable for the intended application. For the GATOR panel, the applicable range of strains is determined by first running the shell analysis with linear elastic shell stiffness (as described in section 2.4) and extracting the maximum and minimum strains seen by the model throughout its deformation. These values are presented in table 3.

Note that the range is asymmetrical in ϵ_{xx} to include the strains seen by the model in cases with applied pre-tension along the span, and to provide a finer resolution for the stiffness data under compression, as the GATOR panel is significantly more compliant in compression than tension. The tension in the GATOR panel is reacted through a combination of chevron wall bending and facesheet stretching, whereas, in compression, the facesheets will tend to buckle (depending in part on the amount of pre-strain applied), after which they do not provide significant axial stiffness. The strain states sampled have 100 linearly distributed points along each of the strain components, as well as for every ordered pair of strain components, considering all permutations, where the two components are varied with equal magnitudes. The latter consideration is essential to capture the material response under combined (bi-axial/bi-directional) loading. Although the dimensionality can be easily expanded to consider triaxial or responses of even higher order for better generality, it is not considered here, as computational resources are the limiting factor.

The complete database consists of ***ABD*** matrices as a function of 6 unique strain components and 15 strain pairs, totalling $(6 + 15) \times 100 = 2100$ simulated data points. For each row of data, 7 simulations are required to populate the ABD matrix, i.e. 1 simulation for the unit cell response under the prescribed strains and 6 simulations to apply small perturbations along each strain axis. The unit cell has 7,232 elements with a total of 41 490° of freedom. The computation time for the 7 simulations and subsequent homogenisation averages at 10 min, depending on the degree of nonlinearity in the system. By batch-running the unit cell models on 20 nodes in a cluster, the complete database was generated in 17 h.

Several methods are available for fitting surrogate models to a database, including polynomial regression, kriging, artificial neural networks, and radial basis functions, among others. Considering the dimensionality of the shell problem, the number of data points, and the readily available software infrastructure, an artificial neural network is employed in this work. The feature space consists of the 6 strain components, and the targets are the 18 unique ***ABD*** components. The mean absolute error (MAE) is used in the loss function and also as a monitored metric, where training is stopped when the metric shows divergence or no convergence within 5 epochs (iterations).

As the targets in the ***ABD*** matrix differ significantly in magnitude, the data for each target is first normalised to the 0–1 range before training, with 0 corresponding to the minimum and 1 to the maximum. Hence, during inference, the predicted values are scaled (i.e. inverse normalisation) using the minimum and maximum values for each target in the training dataset. In the neural network model, a layer normalisation is applied to the input layer to improve numerical stability. The mean and variance values for layer normalisation are tunable parameters that are learnt from the data during training [48]. The hidden layers use rectified linear unit as the activation function. A dropout layer is used as the penultimate layer with a 5% dropout rate for regularisation. The output layer uses the sigmoid activation function. The model is trained using the adaptive moment estimation optimiser [49].

Typically, the optimal feedforward neural network architecture that provides the best fit to a database is not known *a priori*, due to the difficulty in interpreting the neuron coefficients. Here, the same database is used for training networks with different architectures, i.e. wide, shallow, narrow, and deep. In addition, k-fold cross-validation, with $k = 5$, is also used, whereby all the subsamples are used once for validation during training to prevent overfitting. In total, 30 networks are trained, and the top 5 performing networks are saved for the surrogate model. An ensemble average approach is used for model inference, which means that during model runtime, the ABD matrix used in the shell element is not derived from a single network but rather from the averaged outputs of the top 5 performing networks.

In terms of implementation, the fairing models are represented as S4R shell elements in Abaqus, which can account for large rotation and finite strains. The simulations were run using the dynamic implicit solver, providing quasi-static solutions. Convergence was achieved for all cases, and no special treatment was required in the solution process. The nonlinear shell material model for GATOR is introduced through the user subroutine UGENS, which is called at every integration point. The neural networks are created and trained using TensorFlow in Python. The TensorFlow models are directly interfaced from within an Abaqus user subroutine (natively in FORTRAN) using the TensorFlow C interface API—this is achieved through an open-source library (fortran-tf-lib) [50].

3. Results and discussion

The dimensions of the GATOR panel geometric variables are presented in table 1. The material properties of the core and facesheet in the GATOR panel are shown in table 2. The GATOR panels, defined by these dimensions and material properties, are studied using the methods described in section 2. This section represents the results of the analysis organised to assess the effects of various modelling assumptions.

3.1. Timoshenko beam assumption

This study compares the error introduced by modelling the core walls as Timoshenko beams to evaluate the equivalent core stiffness. The original formulation from Olympio and Gandhi [6] and its modified formulation presented in this work are compared to the FE-based homogenisation of the core. The definition of the core dimensions used by Olympio and Gandhi = [6] = [6] is shown in blue in figure 2. The modified formulation employs the shortened effective length for the chevron and the rib height, as shown in red in figure 2. The equivalent stiffness properties evaluated for the core using these methods are shown in table 4, along with the error calculated relative to the FE-homogenised values.

Both analytical formulations assume Poisson's ratio to be zero, and the FE-based homogenisation predicts a negligible value for it, thereby yielding a close agreement between the methods. Similarly, the equivalent E_2 evaluated from all the methods are similar, giving a maximum error of ±4%. However, significant errors are found for the equivalent E_1 and G_{12} values with the original formulation. Axial

Table 4. Comparison of analytical and FE-based homogenisation for the default configuration.

	FEM	Olympio and Gandhi (2009)	Modified formulation
ν_{12}	−0.001	0	0
E_1 (Pa)	108 946	55 751	115 247
E_2 (Pa)	32 586 859	31 680 000	33 971 448
G_{12} (Pa)	27 390	19 295	29 226
E_1 error		−49%	6%
E_2 error		−3%	4%
G_{12} error		−30%	7%

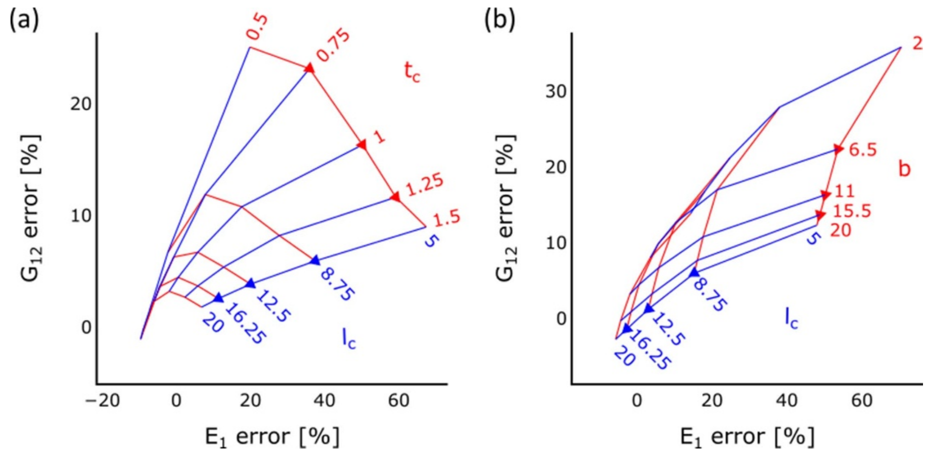

Figure 7. Effect of changing the chevron dimensions on the error between the equivalent moduli values from the modified analytical formulation and the FE-based homogenisation. Note that each carpet plot varies the length of the chevron wall and one of the cross-section dimensions, thereby changing the slenderness ratio of the chevron wall.

deformation of the core in the 1-axis direction requires the bending deformation of chevrons, while in-plane shear deformation requires the bending deformation of both chevrons and ribs. As the effective bending lengths of these walls are less than the length used in the geometry definition, the original formulation significantly underpredicts the core's equivalent E_1 and G_{12} values. The modified formulation uses shortened effective lengths for the chevron wall and rib height, thereby effectively reducing the error between its equivalent moduli values and FE-homogenised values.

The effects of changing the chevron length, thickness and width on the error in the core's equivalent E_1 and G_{12} values evaluated by the modified analytical formulation are presented in figure 7.

The blue lines represent the effects of changing the chevron length (l_c) for constant values of chevron thickness (t_c) in figure 7(a) and for constant values of chevron height (b) in figure 7(b). It shows that for any chevron thickness (t_c) or height (b), increasing chevron length (l_c) reduce the error in the core's equivalent E_1 and G_{12} values. This reduction in error is attributed to the increasing length-to-thickness ratio of the chevron walls. The increasing length-to-thickness ratio of the chevron walls reduces the significance of chevron deformation modes that are not accounted for in the analytical formulation, such as thickness-wise and height-wise deformation due to Poisson's ratio. Hence, the equivalent moduli values from the modified analytical formulation get closer to the FE-homogenised values.

The red lines show the effects of changing chevron thickness (t_c) in figure 7(a) and the effect of changing chevron height (b) in figure 7(b) for constant values of the chevron length (l_c). These red lines in figure 7(a), indicates that error in equivalent E_1 decreases with decreasing chevron thickness (t_c). Similar to increasing chevron length (l_c), this trend is attributed to the increasing length-to-thickness ratio of the walls which reduces the significance of the deformation modes not accounted for in the analytical model. In contrast to the trend in equivalent E_1, the error in equivalent G_{12} initially increases with decreasing chevron thickness (t_c) before reversing the trend for further decrease in chevrons thickness (t_c) for the cases with long chevrons (l_c). While the latter decrease in the error is attributed to the increasing length-to-thickness ratio of the beams, the initial increase in the error with decreasing chevron thickness (t_c) is unintuitive. Hence, it should be explored in further analytical studies.

Another unintuitive trend is shown by the red lines in figure 7(b), where the error in equivalent E_1 and G_{12} values increase with decreasing chevron height (b). The decreasing chevron height (b) increases

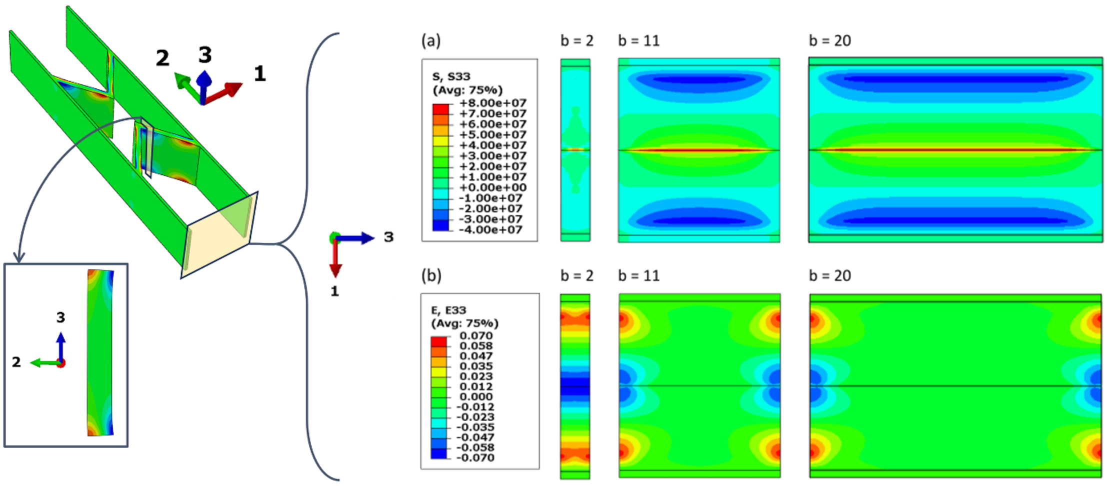

Figure 8. Micro-scale direct stress (σ_{33}) and strain (ε_{33}) distribution in the height-wise (3-axis) direction for various core heights for the load case of $\varepsilon_{xx} = 1$ in the FE-based homogenisation. (a) Shows the stress distribution, and (b) shows the strain distribution.

the length-to-height ratio of the chevron walls, thereby making the wall shape closer to a typical beam rather than a plate despite the increasing error. In this case, the increasing error between the analytical and the FE-based methods stems solely from the changes in equivalent moduli values in the FE-based homogenisation, while the analytical values remain constant. This change in FE-homogenised values is attributed to the change in height-wise (3-axis) strain distribution as the chevron wall deforms in bending, as shown in figure 8.

As the chevron walls bend, they develop significant length-wise strains on the free lateral faces, with tensile strain on one side and compressive strain on the other. These length-wise strains on the faces further cause height-wise strains due to the high Poisson's ratio of the core material (i.e. 0.48, as shown in table 2), resulting in an anticlastic curvature. These height-wise strains become significant as the core height decreases, as shown in figure 8(b). These height-wise strains introduce an additional deformation mode not accounted for in the analytical models, thereby increasing the error as this deformation mode becomes more significant with decreasing core height (b).

The results of comparing the equivalent moduli from the modified analytical formulation and FE-based homogenisation show the limitations of the analytical approach in accurately calculating the equivalent moduli for a variety of core geometries. The analytical approach shows good agreement with FE-based homogenisation for cases with a high length-to-thickness ratio for walls and an adequate height to reduce the effects of height-wise strains. However, the analytical approach is not suitable for a parametric study of the core dimensions, as it may introduce significant errors for certain geometries.

3.2. Laminated plate assumption

The equivalent shell stiffness of the GATOR panels is evaluated using an analytical and FE-based homogenisation method, as described in sections 2.1 and 2.2, respectively. Additionally, a combined approach, in which FE-homogenised core properties are used with the CLPT to evaluate the equivalent shell stiffness of the GATOR panel, is applied to two cases. In the first case, only the in-plane properties of the cores are used to evaluate both the ***A*** and ***D*** matrices. In the second case, both in-plane and out-of-plane equivalent properties are used to evaluate the ***A*** and ***D*** matrices, respectively. The comparison of the equivalent shell stiffness properties evaluated for the GATOR panel by different methods, as shown in figure 9, helps isolate the effects of individual modelling simplifications and assess their impact on the accuracy of the homogenised stiffness properties.

The modelling simplifications presented by bars of each colour in figure 9 are summarised as follows:

(1) Red bars—homogenisation of core properties using the analytical method as opposed to the FE-based method.

(2) Orange bars—usage of In-plane properties of the core to evaluate the ***D***-matrix as opposed to using the out-of-plane properties of the core.

(3) Blue bars—negligence of the core-facesheet interaction as opposed to accounting for it.

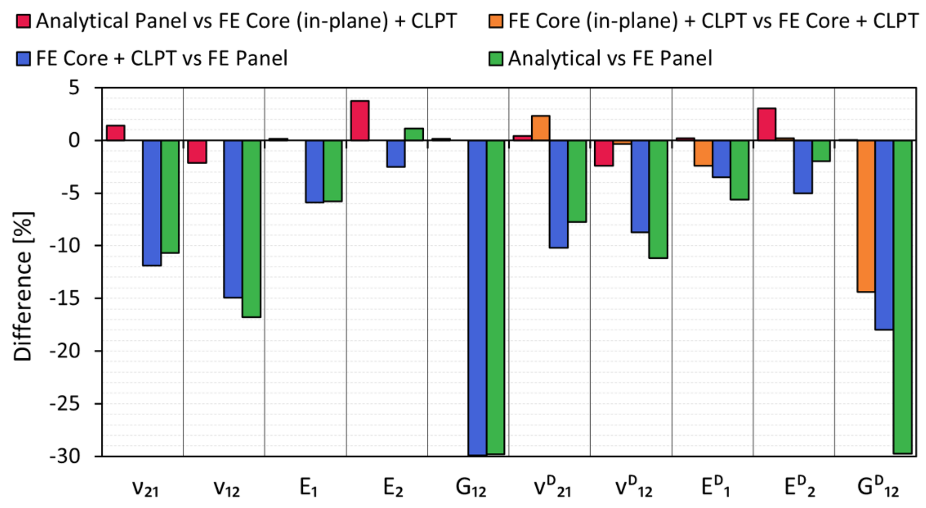

Figure 9. Comparison of various methods of homogenising the GATOR panel stiffness properties.

(4) Green bars—usage of fully analytical homogenisation method as opposed to fully FE-based homogenisation method.

The red bars compare the stiffness values from the modified analytical method with those from the combined method using only the in-plane properties of the core. It shows close agreement with an error of less than ±4% due to similar core properties from both methods, as presented in table 4. The figure further shows that the error in the GATOR panel's equivalent E_1 and G_{12} becomes negligible with an error of less than ±0.2%. The facesheet properties are identical in the two methods, while the core properties differ by up to 7%, as shown in table 4. Hence, the negligible error in the GATOR panel's equivalent E_1 and G_{12} indicate that the predominant stiffness contribution to these two deformation modes comes from the facesheets rather than the core.

Cellular cores typically have vastly different equivalent moduli for in-plane and out-of-plane deformation modes. Hence, the GATOR panel's equivalent properties evaluated by the combined method using only the in-plane moduli of the core are different from the combined method that uses both in-plane and out-of-plane moduli of the core. The comparison of these two approaches is shown by the orange bars in figure 9. It shows that the in-plane properties are identical between the methods, while the out-of-plane properties are significantly different, particularly for equivalent E_1^D and G_{12}^D. The negative error bars for equivalent E_1^D and G_{12}^D indicate that these out-of-plane moduli of the core are greater than the corresponding in-plane moduli. Especially the error of −14% for the equivalent G_{12}^D indicates that the torsional stiffness of the core is significantly higher than its in-plane shear stiffness. Hence, to improve the accuracy of the evaluated equivalent shell stiffness for the GATOR panel, the out-of-plane moduli of the core should be explicitly evaluated.

The blue bars in figure 9 show the error introduced by ignoring the core-facesheet interaction in the GATOR panel. The blue bars compare stiffness properties from the combined method, which uses FE-homogenised core properties in the CLPT, against those from the FE-based homogenisation of the GATOR panel unit cell. Hence, the only difference between the two methods is due to the omission of the non-uniform facesheet deformation in the combined method. The figure shows large errors for equivalent Poisson's ratios and moduli, indicating that the effect of core-facesheet interaction on the equivalent stiffness is significant. This error is particularly substantial for equivalent G_{12} and G_{12}^D with values of −30% and −18%, respectively. Hence, the omission of core-facesheet interaction significantly underpredicts the equivalent stiffness of the GATOR panel.

The green bars in figure 9 compare the equivalent stiffness from the fully analytical homogenisation method to that from the fully FE-based homogenisation method. The analytical homogenisation ignores the core-facesheet interaction and uses the core's in-plane stiffness to evaluate the GATOR panel's in-plane and out-of-plane properties. Hence, the green bars show significant errors due to the simplifications in the analytical model, giving errors as large as −30% for equivalent G_{12} and G_{12}^D. Hence, for improved accuracy of the homogenised stiffness properties, the effects of the out-of-plane stiffness of the core and the core-facesheet interaction must be captured by the homogenisation method used for the GATOR panel.

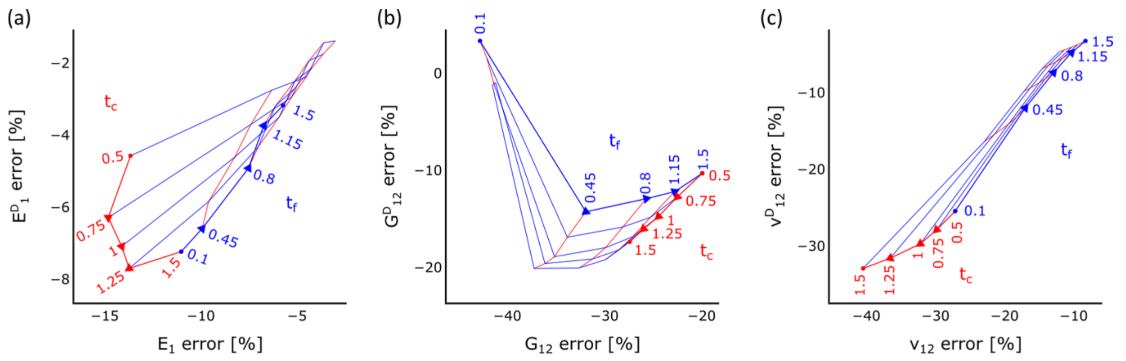

Figure 10. The error in equivalent properties from the combined approach, using FE-homogenised core properties in classical laminated plate theory, relative to FE-based homogenisation of the GATOR panel. It shows the error due to facesheet-core interaction for various values of facesheet and chevron thickness.

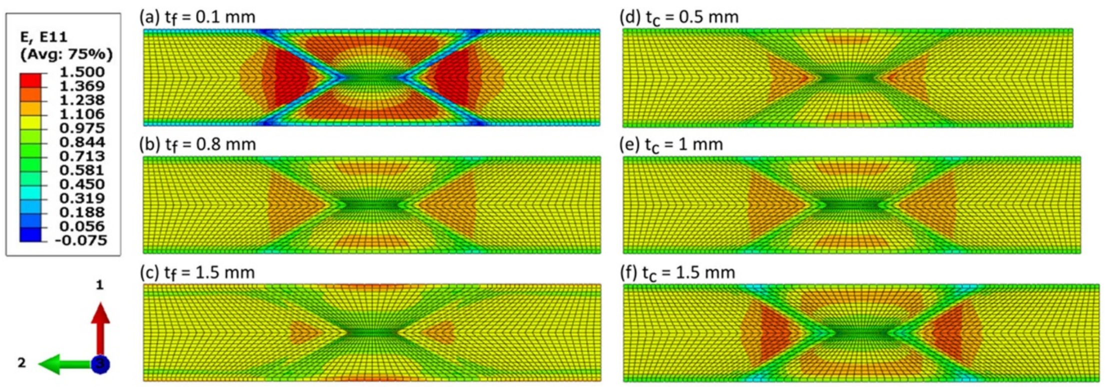

Figure 11. The micro-scale direct strain distribution (ε_{11}) in the morphing direction (1-axis) on the facesheet from the top view for an applied macro-scale strain of $\varepsilon_{xx} = 1$ for various values of facesheet and chevron thickness.

The effects of core-facesheet interaction are further studied by varying the chevron thickness (t_c) and facesheet thickness (t_f). In this study, the equivalent stiffness obtained from the combined method, which incorporates FE-homogenised core properties into the CLPT, is compared with that from FE-based homogenisation of the GATOR unit cell. The errors in the equivalent GATOR panel properties are shown in figure 10, and the non-uniform strain distributions on the facesheet are shown in figure 11.

The blue lines in figure 10 show decreasing errors for increasing facesheet thickness (t_f), except for the anomaly in the very thin facesheets (i.e. $t_f = 0.1$) for out-of-plane shear modulus G^D_{12}. The anomaly indicates that the effect of facesheet-core interaction in torsional deformation is negligibly small for very thin facesheets compared to facesheet thickness (t_f) above 0.45 mm. Hence, a similar equivalent G^D_{12} is evaluated from the combined method and the FE-based homogenisation. The reduction in error with increasing facesheet thickness (t_f) indicates that the non-uniformity of the facesheet deformation due to facesheet-core interaction decreases with increasing facesheet thickness (t_f). This decrease in non-uniformity of the facesheet deformation is shown in figures 11(a)–(c) for axial deformation of the unit cell (i.e. $\varepsilon_{xx} = 1$). It shows that increasing facesheet thickness (t_f) reduces the ability of the chevrons to pull the facesheet in the direction of their displacement, thereby reducing non-uniformity in the facesheet deformation.

Similarly, the red lines in figure 10 show decreasing errors for decreasing chevron thickness (t_c), except for the anomaly in the very thin facesheets (i.e. $t_f = 0.1$). Decreasing chevron thickness (t_c) reduces the stiffness of the chevron walls. This reduction in the chevron stiffness reduces their ability to pull the facesheets in the direction of their displacement. Hence, decreasing chevron thickness (t_c) leads to reducing non-uniformity in facesheet deformation, as shown in figure 11(d) to figure 11(f). The decreasing non-uniformity in the facesheet deformation reduces the effects of core-facesheet interactions, making the equivalent stiffness from the CLPT closer to that from the FE-based homogenisation of the GATOR unit cell.

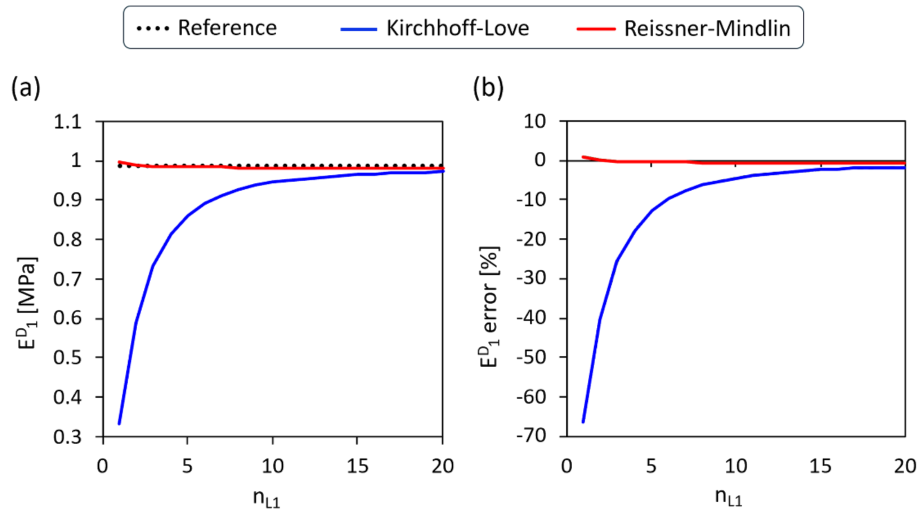

Figure 12. Equivalent flexural modulus (E_1^D) along x -axis evaluated by fitting full-scale FE analysis data to Kirchhoff–Love and Reissner–Mindlin plate equations. The reference value is from the FE-based homogenisation of the GATOR panel unit cell. (a) Shows the equivalent E_1^D values, (b) shows their error relative to the reference value.

These results comparing the equivalent stiffness from CLPT and FE-based homogenisation show that neglecting core-facesheet interaction introduces significant errors in the equivalent stiffness properties of the GATOR panel. The CLPT results with the core's in-plane properties alone compared against those with the core's in-plane and out-of-plane properties show that the negligence of the core's out-of-plane properties also introduces significant error, particularly in the torsional stiffness of the GATOR panel. Hence, the analytical approach to homogenising the equivalent stiffness of the GATOR panel is inadequate for the multiscale modelling of the GATOR panel fairing for folding wingtips.

3.3. Kirchhoff–Love plate assumption

The FE-based homogenisation method proposed for the multiscale modelling approach does not evaluate the equivalent transverse shear stiffness. While transverse shear deformations can be significant for panels with a low length-to-thickness ratio, their effect decreases as the panel length increases. Hence, this section examines the effect of transverse shear deformation for various lengths of the GATOR panel using a cantilevered panel with tip displacement loading, as described in section 2.3. The displacements and the reaction loads from the full-scale FE simulations are used to evaluate an equivalent flexural modulus (E_1^D). The equivalent E_1^D values from the study are shown in figure 12, along with their error relative to FE-based homogenisation values.

Figure 12(a) shows that for short lengths of the panel, the equivalent E_1^D from the Kirchhoff–Love plate model is significantly lower relative to the reference value from the FE-based homogenisation, thereby giving a significant error in figure 12(b). This underprediction of equivalent E_1^D is due to the transverse shear deformation mode that is neglected in the Kirchhoff-Love plate model. It further shows that as the number of cells along the length (i.e. x-axis) of the panel increases, the equivalent E_1^D from the Kirchhoff–Love plate model gets closer to the reference value. This reduction in the error is attributed to the increasing length-to-thickness ratio of the panel, which reduces the significance of the transverse shear deformation. In contrast, the Reissner–Mindlin plate model accounts for the transverse shear deformation. Hence, the equivalent E_1^D from the Reissner–Mindlin plate model shows close agreement with the reference value.

The error from the Kirchhoff–Love formulation is smaller than −5% for 10 cells and smaller than −3% for 14 cells. The error from the Reissner–Mindlin formulation is always smaller than −0.7%. The unsupported length of the GATOR panel in the target problem of wing fairing is 16 cells. This is the length between the ribs in the wing section shown in figure 1. At this panel length, the error in the equivalent E_1^D from the Kirchhoff–Love plate model is smaller than −2.3%. For the analysis of large morphing structures, such as the target problem discussed, this error is negligible, particularly given the need for a scalable modelling approach and the computational cost of evaluating the transverse shear stiffness in FE homogenisation, as discussed in Section 1.3. Hence, the shell formulation based on Kirchhoff–Love plate theory is used to model the GATOR panel fairing on the wing section.

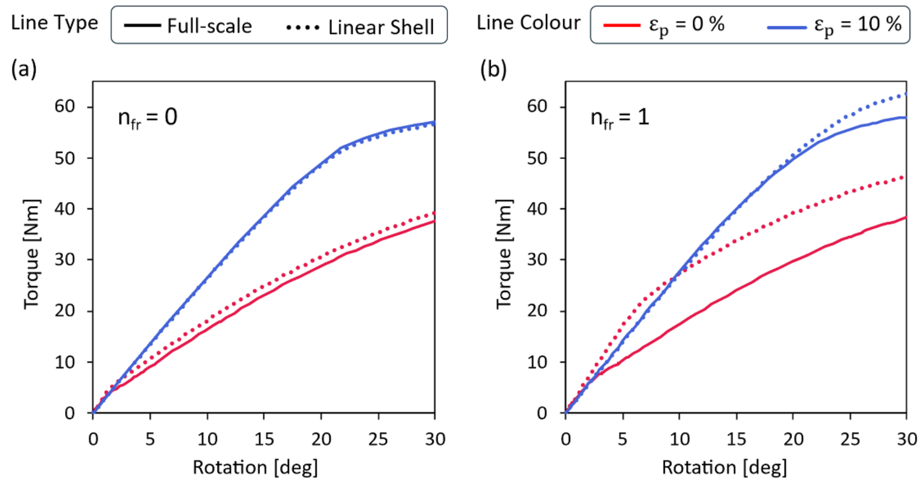

Figure 13. Comparison of torque response between multiscale and full-scale for the twin-plate models.

3.4. Linear elasticity assumption

This section examines the effects of geometric nonlinearity in the GATOR panel deformation, which alters its equivalent stiffness properties as the panel deforms. For the shell fairing model with linear stiffness, the GATOR panel stiffness properties are homogenised at the undeformed state, and these homogenised properties are used to define the linear elastic constitutive properties of the shell model. Hence, the effects of geometric nonlinearity in the GATOR panel deformation are not captured by this shell model. This section evaluates the error introduced by this shell model by comparing it to the full-scale model for a simplified fairing slice geometry described in section 2.4. The torque-rotation responses from the two models for various values of applied pre-tension and floating ribs are shown in figure 13.

As shown in figure 13(a), there is close agreement between the shell and full-scale model responses for the fairing slice without a floating rib over the full 30° rotation range. The error in the shell model's torque response at 30° rotation is 3.9% and −0.9% for the case with 0% and 10% pre-strain, respectively. The case without pre-strain has a slightly higher error than the case with pre-strain due to more prominent local buckling of the facesheet on the top panel. The error due to the local buckling of the facesheet is more significant in cases with a floating rib, as shown in figure 13(b). It shows that the torque response of the shell model deviates significantly from that of the full-scale model in both cases, but at different rotation angles. For the case without pre-strain, the shell model's torque response deviates (with an error exceeding 10%) from the full-scale model after about 2° of rotation. The case with 10% pre-strain shows close agreement between the shell and full-scale models up to a rotation of 20°, after which the shell model's response deviates from the full-scale model's response.

The trends in the torque response are explained through various features in the deformed shapes of the full-scale fairing slices shown in figure 14 for all configurations at approximately a 20° rotation angle.

It shows that the cases without pre-strain in both configurations have more prominent buckling of the facesheet than those with 10% pre-strain. Comparing the two cases without pre-strain in figures 14(a) and (c) indicates that the case with a floating rib has localised regions with steep panel curvatures that have the cells with buckled facesheets, whereas in the case without floating ribs, these regions are more spread out and have shallower panel curvatures. Moreover, the amplitudes of buckled facesheets are larger in the case with a floating rib compared to the case without it. These comparisons indicate that the case with a floating rib creates more prominent changes in the cell geometry of the GATOR panel as the pivoting rib is rotated. As the stiffness changes due to geometric nonlinearity in GATOR panel deformation are not captured by the shell model, it shows significant deviation from the full-scale model for the cases with a floating rib.

This study indicates that the shell model with linear stiffness and full-scale models show good agreement in their torque response up until the rotation angle at which the facesheets in the full-scale model undergo significant local buckling. It further shows that the applied pre-tension on the panel delays the rotation angle at which the facesheets undergo local buckling, thereby extending the rotation angles up to which both models give similar results. However, it also highlights that the shell model with linear

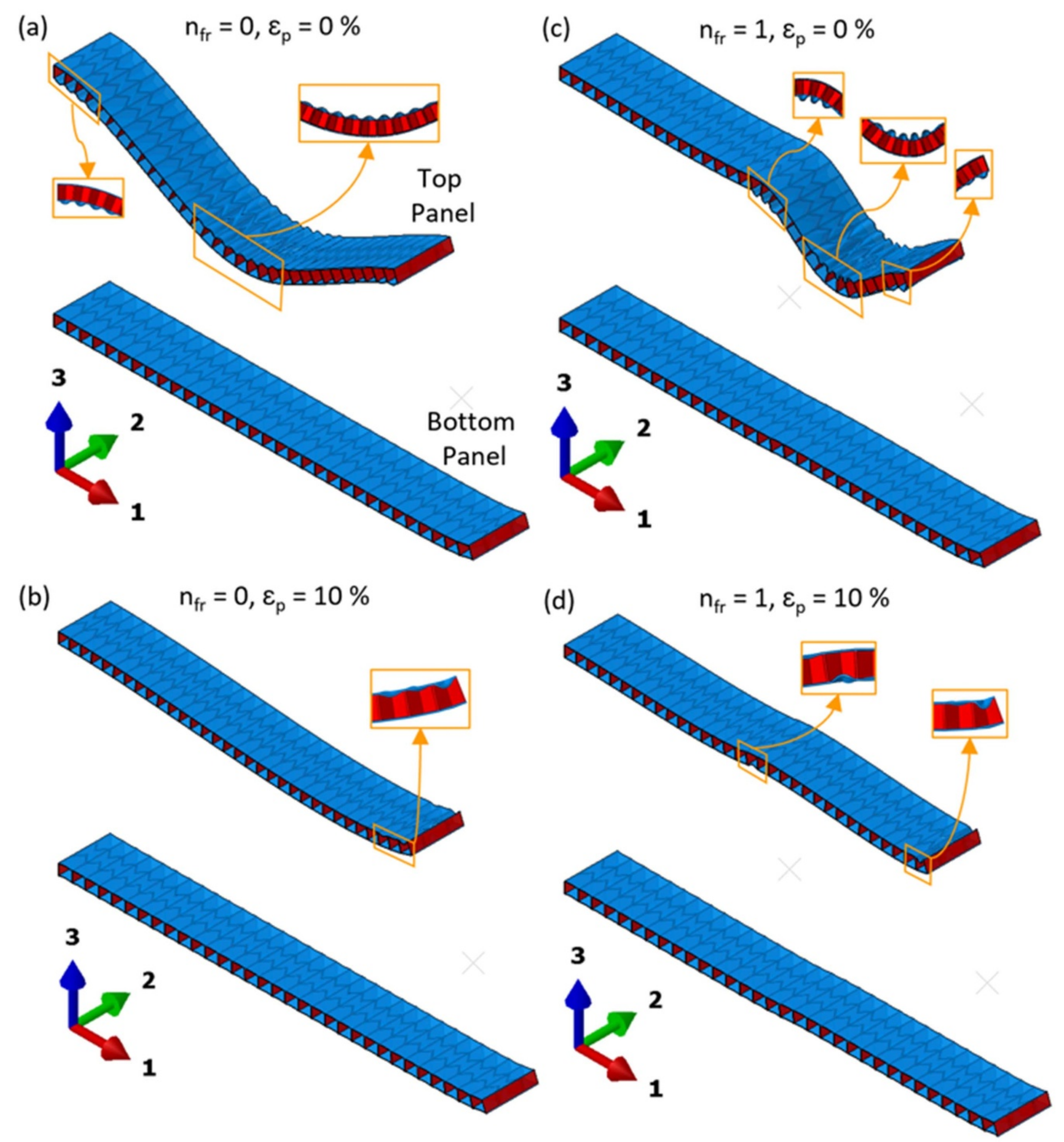

Figure 14. Deformed shapes of the full-scale model of the fairing slice at 20° rotation (a) and (b) show the 0% and 10% pre-strain case for the fairing slice without floating rib. (c) and (d) show the 0% and 10% pre-strain case for the fairing slice with a floating rib.

stiffness is inadequate for accurately representing the GATOR panel fairing response in cases with significant localised deformation. As an alternative approach to address this limitation, the following section presents a computationally efficient methodology based on neural networks for capturing the nonlinear stiffness response of the GATOR panel in the shell-based representation of the fairing.

3.5. Data-driven multiscale modelling results

In this section, the data-driven multiscale shell model is used to simulate the fairing slice. As described in section 2.5, this model bridges the geometric nonlinearity in the GATOR panel responses to the shell model of the fairing by updating the constitutive properties of the shell element as it is deformed. These constitutive properties are the averaged outputs of the top 5 performing neural networks, whose architectures and MAE are shown in table 5.

The notation $A \rightarrow [B] \times C \rightarrow D$ is read as a neural network with A input features, with C hidden layers where each layer has B neurons and D number of outputs. Table 5 shows an average MAE of 0.7%, indicating that the network predictions of the $\boldsymbol{ABD}$ matrix are accurate. The membrane and bending stiffness emulated by the neural networks are compared against the simulated data in figure 15.

Table 5. Mean absolute error (MAE) for top-performing neural network architectures.

Architecture	Mean absolute error (MAE)
$6 \rightarrow [180] \times 6 \rightarrow 18$	0.006 43
$6 \rightarrow [180] \times 6 \rightarrow 18$	0.006 99
$6 \rightarrow [60] \times 6 \rightarrow 18$	0.007 52
$6 \rightarrow [60] \times 4 \rightarrow 18$	0.007 53
$6 \rightarrow [60] \times 6 \rightarrow 18$	0.007 57

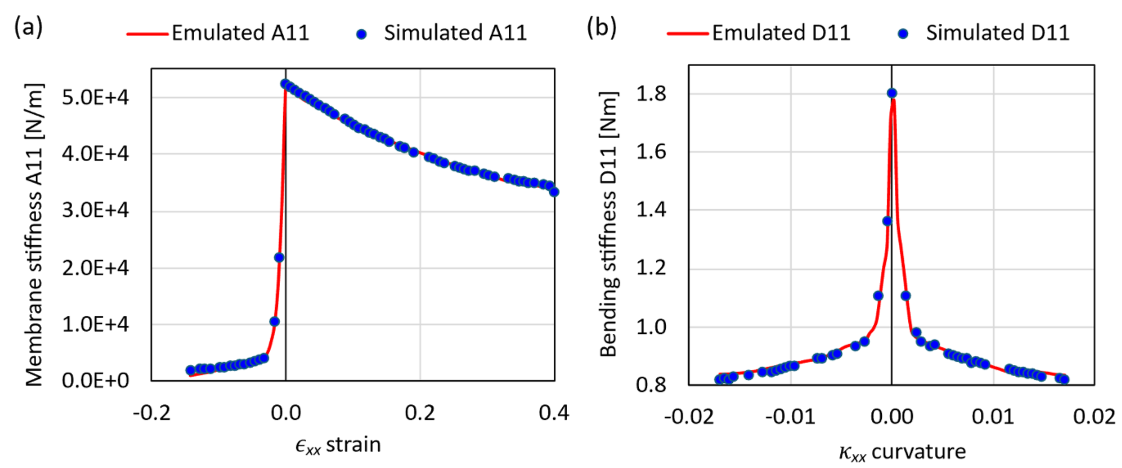

Figure 15. Emulated membrane and bending stiffness in the x-axis compared against the simulated data.

It shows that the neural network can accurately emulate the effective material response of the panel despite the sharp changes in the trends. Figure 15(a) elucidates the high degree of geometric nonlinearity in the GATOR panel, where there is a very rapid change in the membrane stiffness under initial compression. This reduction plateaus at −3% strains, with a 90% reduction relative to the linear elastic value. In the tension region, the membrane stiffness reduces more progressively, with a 30% reduction at 40% strain. This softening is attributed to the thinning of facesheets as they are stretched, resulting in a reduced incremental increase in the reaction force that resists deformation. From figure 15(b), it can be seen that the panel shows the same response under positive and negative curvature strains. This symmetry in the curvature response is expected as the panel is symmetric about all principal Cartesian planes. Interestingly, the bending stiffness also reduces drastically with the bending strains due to the buckling of the facesheets on the compression side, which diminishes its ability to carry loads.

Three cases are studied here using the fairing with 1 floating rib for pre-tensions at 0%, 5% and 10% strains, respectively. The multiscale model results are benchmarked against high-fidelity solutions, where the fairing is fully discretised with solid FEs. The results from this data-driven shell model, the previously presented linear shell model and the full-scale model are shown in figure 16.

It shows that the data-driven shell models are capturing the overall trend in the global responses, where the moment increases linearly with the rotations up to a bifurcation point, after which the torque-rotation curve softens. The bifurcation points for the 0%, 5%, and 10% pre-tension cases are near 2, 12 and 22°, respectively. The data-driven shell models accurately capture the initial responses in all cases where the structural response is seemingly linear. However, this is not strictly the case. As shown in section 2.5, the membrane stiffness reduces by nearly 8% at 5% tensile strain and 16% at 10% strain. The data-driven shell model and full-scale model solutions diverge after the bifurcation point, with discrepancies maintained within 5% of the reaction moment. This discrepancy is likely due to the strain localisation near the floating ribs—the shell response is stiffer as the first-order shear models cannot account for the high degree of cross-sectional distortions that are present in the high-fidelity models.

The data-driven shell model shows a significant improvement in solution accuracy relative to the shell model with linear constitutive properties. Given that the full-scale model is prohibitively expensive, even for this simplified geometry, which takes an average of 8 h with 10 cores, the data-driven multiscale model offers a suitable alternative at a fraction of the computational cost. Once a surrogate model is generated for the equivalent constitutive properties of the GATOR panel, which takes approximately 17 h, the data-driven shell analysis runs in an average of 2 min with a single core. Hence, the

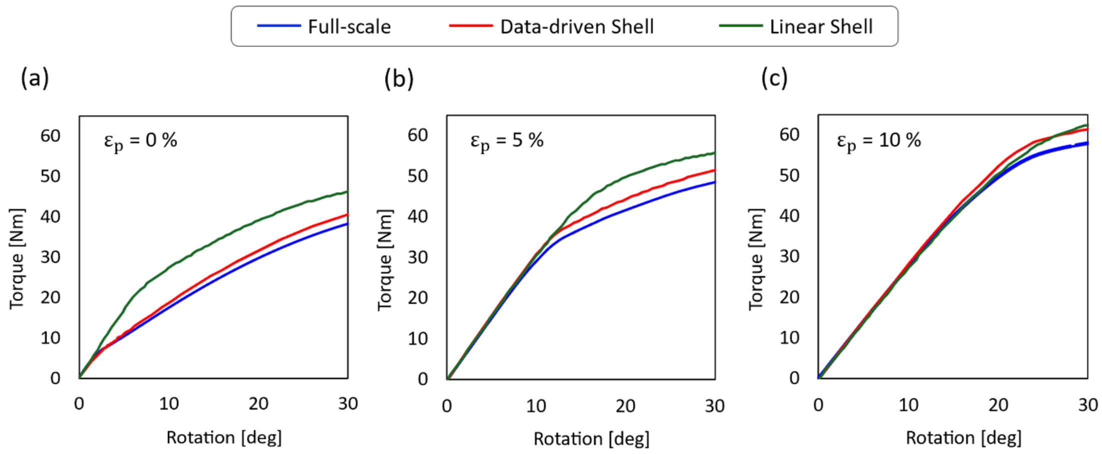

Figure 16. Comparison of data-driven shell model, linear stiffness shell model and full-scale model.

data-driven multiscale modelling offers a computationally efficient approach to accurately modelling the GATOR panel response as a fairing for folding wingtip joints.

4. Conclusions

This paper proposed a multiscale modelling approach as an alternative to the full-scale, full-fidelity modelling of the GATOR panel fairing for folding wingtip joints. The multiscale modelling approach comprises two steps: (1) homogenisation of the GATOR panel stiffness properties to an equivalent shell stiffness matrix, and (2) shell-based modelling of the fairing with homogenised GATOR panel properties. The equivalent stiffness properties are evaluated using analytical homogenisation or second-order solid-to-shell homogenisation using FE methods. The study showed that analytical models underpredict the core's stiffness, and this can be corrected by using a shortened effective length for the bending members rather than their full length. It also showed that the core-facesheet interaction in the GATOR panel is significant. Hence, the panel's stiffness properties are underpredicted by the CLPT, which does not account for the interaction between the layers. The paper further showed that the effect of transverse shear decreases with the length of the panel and that its effect becomes negligibly small for the unsupported panel length present in the wing section geometry. Hence, the FE-based homogenisation approach is used in the multiscale model to evaluate the required stiffness properties for the shell model based on the Kirchhoff plate theory. The comparison of the linear elastic shell model and full-scale model showed good agreement in the torque response up to the point where the GATOR panels undergo significant localised deformation in the form of facesheet buckling. In contrast, the effects of geometric nonlinearity in the GATOR panel deformation became significant at large rotations of the fairing slice, particularly in cases with floating ribs. The error introduced in the multiscale models due to the geometric nonlinearity in the GATOR panel deformation was reduced by using a data-driven surrogate model, which updated the constitutive properties of the shell elements as they deformed. This data-driven multiscale model showed close agreement with the full-scale model for large rotation angles, while significantly reducing the computational cost relative to the full-scale model. Hence, the data-driven multiscale model offers a scalable and cost-effective approach to studying uniform GATOR panels as fairings for folding wingtip joints. Future research on the data-driven modelling approach should further reduce the cost of generating the surrogate model to model spatially non-uniform morphing panels in large structures.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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